
Simulation Testbed for a Power System with a High Penetration of Grid-Forming Inverters

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Executive Summary

We developed a reusable simulation testbed to study power systems with a high share of generation provided by grid-forming (GFM) inverters. The University of Texas at Austin collaboration with the Electric Reliability Council of Texas (ERCOT) began with an IEEE 39-bus PSS/E network, translated to PSCAD through the E-TRAN software package. We equipped the testbed with machine dynamics supplied by ERCOT and inverter models from Pacific Northwest National Lab (PNNL) for machine replacement studies. The first-year work primarily concerns electromagnetic transient (EMT) simulations in PSCAD; the PSS/E starting case provides the network foundation for corresponding positive-sequence studies in the next phase of the collaboration.

Battery storage motivates this work. Batteries absorb energy when supply is abundant and can inject it on short notice. Variable renewable generation and rapidly changing loads raise the value of that flexibility. ERCOT's advanced grid-support requirements for new inverter-based storage resources also make GFM behavior increasingly relevant to system studies. Understanding how these controls behave in steady state, during faults, and in the post-fault period is therefore important.

The testbed combines configurable network disturbances, resource models, and scripts that process the simulation outputs. Eleven configurations replace zero through ten synchronous machines with the selected droop-controlled GFM model; the testbed also supports later studies of other GFM controls, grid-following (GFL) resources, mixed generation, user-defined load profiles, and load disturbances. The results are specific to this model and the tested cases: one droop-controlled GFM implementation and a limited set of disturbances at two buses within the 39 bus network. Other GFM controls, other settings of this one, or other scenarios may behave differently.

The reference simulations show fault-ride-through recovery of the connected network in generation configuration from all-machine systems to all-GFM systems. The selected GFM responses settle faster than the corresponding machine responses after the three-phase to ground, "bolted" fault.

The analytical comparison connects these observations to familiar machine dynamics. A synchronous machine's rotor motion sets its electrical phase. A GFM controller generates that phase without a rotating shaft. Under unsaturated filtered droop, the two models can be written in the same reduced swing-equation form. The GFM droop gain sets an equivalent damping coefficient $D_{eq} = 1/m_p$, while the power-measurement filter and droop gain set equivalent inertia. For the selected parameters, the low-frequency damping coefficient is five times the machine governor droop coefficient on the corresponding normalized bases.

This equivalent damping explains the strongly damped GFM responses, but it is not the only control property that affects fault recovery. Current and voltage limits change the power that an inverter can deliver during severe disturbances. The inertia estimation tests further show that damping-related power contributes to the estimated inertia over the test window. The preliminary transient-stability investigation compares critical-clearing-time estimates from classical energy methods with the fault-duration limits found in EMT simulation.

The next phase will align topology, operating points, ratings, bases, applicable controls, and disturbances in corresponding PSS/E and PSCAD studies. The objective is to understand differences between modeling domains. We will also extend the preliminary analytical work to investigate how current limits, voltage limits, and droop settings influence transient stability. Together they provide a foundation for guidance to manufacturers and power system engineers.

1 Introduction

This report presents the first-year PSCAD electromagnetic transient (EMT) testbed and its initial application to machine and inverter disturbance studies. It follows the development of the network, the simulated responses, and the reduced dynamic models used to interpret them. The adapted IEEE 39-bus system provides a common setting for this comparison, while its PSS/E origin supports the next phase of corresponding positive-sequence studies. The Electric Reliability Council of Texas (ERCOT) hosts 21.7 GW of battery energy storage in the reporting-period integration report [2]. Approximately 36 GW of additional storage projects hold signed interconnection agreements [3]. These projects indicate substantial development activity, although an agreement does not guarantee commercial operation. Batteries provide energy to ERCOT customers and can also support frequency regulation and load smoothing.

Artificial intelligence (AI) data centers introduce rapid demand changes during model training and model service. Batteries can reduce the data center power ramps presented to the grid, depending on facility design and control [4]. They may connect directly to the grid or provide on-site power alongside uninterruptible power supplies. These applications motivate understanding the dynamic behavior of the inverter interface through which a battery delivers or absorbs power.

A battery connects to a three-phase AC network through a DC–AC inverter, as Figure 1 illustrates. Semiconductor switches and passive capacitors and inductors shape the AC output from the DC source. A micro-controller changes each switch’s duty cycle, the fraction of a switching period spent on, using measured voltage and current. The resulting commands regulate the electrical variables needed to deliver the selected power output.

An LCL filter places two inductors and a shunt capacitor between the converter and the grid. The controller regulates current through the converter-side inductor and voltage across the filter capacitor, while outer controls govern power and voltage behavior. Different control approaches use this same physical interface differently. This report focuses on grid-forming (GFM) control and compares its response characteristics to those of conventional synchronous machines.

The Universal Interoperability for Grid-Forming Inverters (UNIFI) Consortium defines GFM behavior through an internal voltage phasor that remains approximately stiff on subtransient-to-transient timescales [5]. This characteristic supports an immediate electrical response to changes at the connection point, subject to resource and control limits. It does not mean that voltage magnitude or phase remains fixed indefinitely, or that the inverter behaves as an unlimited voltage source during a severe fault. GFM control takes several forms, including droop and virtual-synchronous-machine implementations [5, 10, 13]; all maintain that internal voltage phasor as the voltage and frequency reference, and the system-level behavior then depends on the remaining controls, hardware, and operating limits.

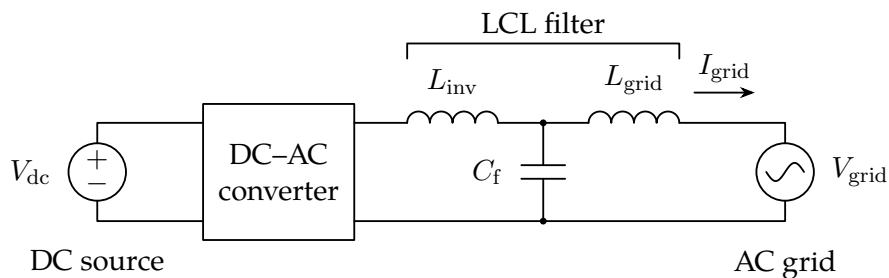


Figure 1. Three-phase inverter interface, shown as a single-phase equivalent: DC source, converter stage, LCL filter, and AC grid.

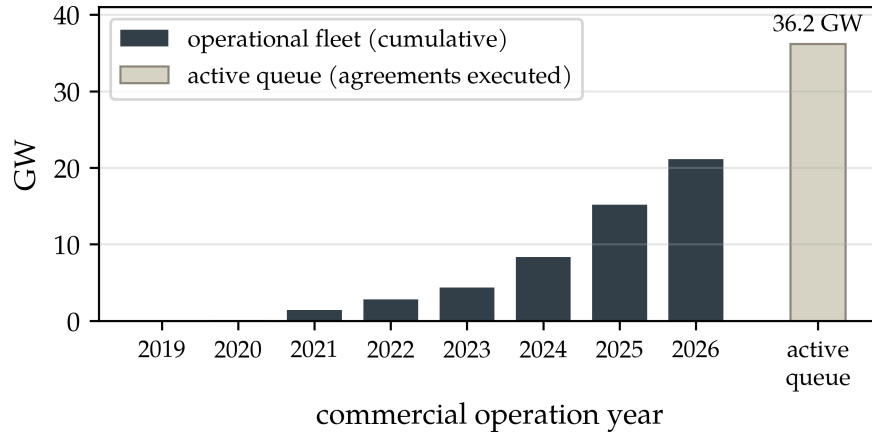


Figure 2. ERCOT battery energy storage capacity by commercial-operation year, alongside 36.2 GW in the interconnection queue with executed agreements. Sources: July 2026 Monthly Outlook for Resource Adequacy and Generator Interconnection Status reports [7, 3].

ERCOT’s Nodal Operating Guide Revision Request (NOGRR) 272 establishes advanced grid-support requirements for inverter-based energy storage resources connected to the ERCOT Transmission Grid [6]. The applicability trigger is a Standard Generation Interconnection Agreement (SGIA) executed on or after April 1, 2026, rather than the date on which any new battery begins operating. The revision was approved on November 6, 2025, with an effective date of December 1, 2025; the SGIA cutoff is a distinct date [6, 39]. This distinction matters when describing the population of resources to which the requirements apply.

Conventional grid-following (GFL) control uses an externally established voltage waveform as its synchronization reference, commonly through a phase-locked loop. Within normal voltage limits, a GFL’s output current phasor remains unchanged immediately after a disturbance, before outer controls adjust the injected power [5]. A solver’s use of a current-source equivalent does not determine this classification: a GFM voltage source behind impedance can also be converted to a Norton equivalent current-source representation. The distinction concerns control behavior and synchronization, not simply the circuit element used in a numerical network solution.

Three terms connect GFM controls to the machine-based interpretation developed in this report. Their meanings depend on the reduced model and its operating assumptions. In particular, an analogy between equations does not make the physical sources identical.

1. The *inverter angle* is a controller-generated electrical phase. Like a machine’s electrical rotor angle, its displacement relative to other sources influences power transfer. It is not the position of a physical shaft.
2. The *equivalent inertia* of filtered droop arises from the lag in its power measurement and depends on the filter time constant and droop gain [1, 19]. A longer filter time constant or a smaller droop gain raises this coefficient, and with it the inverter angle’s resistance to acceleration, without any physical moment of inertia changing. The electrical energy still comes from the DC source and circuit storage.
3. The *equivalent damping* relates normalized frequency deviation to normalized power response. The droop gain sets this coefficient in the reduced GFM model. Section 4.6 distinguishes it from a machine governor’s frequency-dependent contribution.

Section 2 describes the network, resource models, and simulation workflow. Section 3 presents the EMT replacement and fault-response studies. Section 4 develops the machine-to-GFM comparison and interprets the inertia estimation tests. Section 5 presents the preliminary energy analysis and next-phase directions.

2 Simulation Testbed

The testbed provides a configurable network for the machine-to-inverter comparison. We converted the PSS/E case to PSCAD with E-TRAN and used the EMT representation for the first-year disturbance studies. This construction also preserves a common starting point for future corresponding-model comparisons.

2.1 Testbed Construction

The adapted IEEE 39-bus network contains ten generator buses [8, 9]. Its EMT representation includes Bergeron transmission lines, generator step-up transformers, and constant-power loads (E-TRAN load model type 0: fixed real and reactive power) [35]. E-TRAN translates the network parameters and initial load-flow operating point from PSS/E into PSCAD [34]. The load representation matters for the stability estimates of Section 5.1.1: with these constant-power loads the ten-machine estimate is 0.248 s, inside the EMT bracket, whereas a constant-impedance representation of the same loads gives 0.479 s. The resulting benchmark uses a 230 kV transmission network with representative ERCOT-supplied machine dynamics, rather than representing actual Texas transmission facilities.

The ten machines share GENROU, ESST4B, GGOV1, and PSS2B dynamic parameters, including inertia $H = 5.46$ s. Sharing these parameters does not imply identical ratings or dispatch. Figure 5 summarizes the construction and processing workflow. Figures 3 and 4 show the controls used to select fault location, type, timing, and associated breaker action.

The fault controls combine a central configuration panel with logic attached to individual transmission lines. We specify the fault start time, location, type, and duration centrally, while the line controls implement the corresponding fault and breaker commands. The protection settings distinguish removal of the fault from restoration of the connection. This arrangement allows the same network to support either a reclosed connection or a post-fault topology in which the selected connection remains open.

2.2 Iterative Generator Replacement

One design goal was to study mixtures of synchronous machines and GFM inverters in the same network. Each generator location can retain its machine or host an inverter, with the operating point specified in that resource's PSCAD model. We configured the replacements manually for the first study; the PSCAD Python interface (`mhi.pscad`) also provides a path to automated configuration.

We developed eleven configurations by replacing zero through ten machines with GFM units. Because generator ratings and operating points are not identical, replacement count identifies the cases without assuming that each replacement corresponds to exactly 10% of generation capacity. The progression spans the all-machine and all-GFM endpoints while retaining a common reference disturbance. The framework can also accommodate alternative GFM models, machine dynamics, and GFL resources for later mixed-system studies.

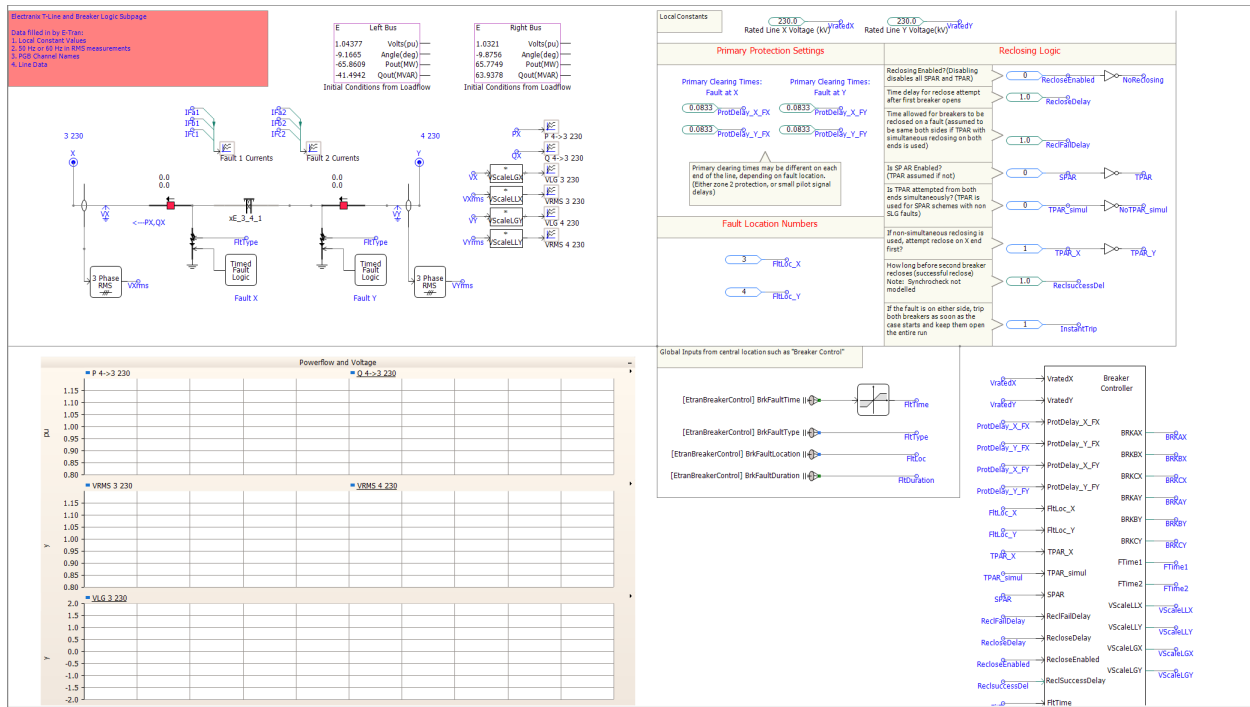


Figure 3. Fault and protection configuration on one E-TRAN-converted line, as presented in the PSCAD user interface: the per-line timed fault logic and measurement channels (left), and the primary protection clearing times, reclosing logic, fault-location numbers, and breaker controller (right).

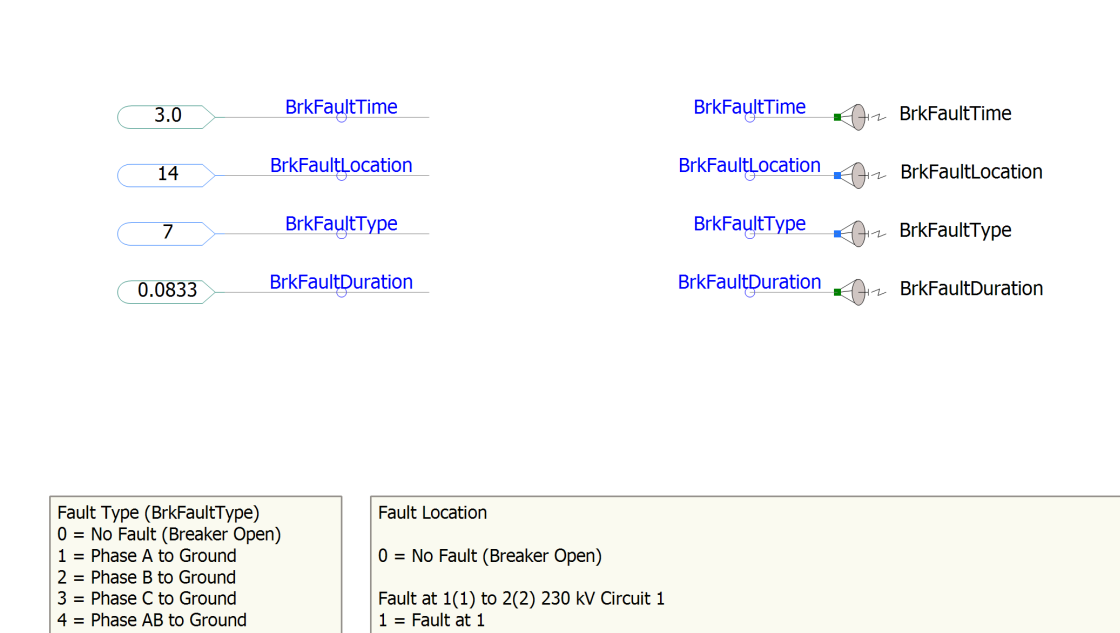


Figure 4. Central fault-control settings for the bus-14 reference disturbance: start time 3.0 s, type code 7 (3 phase to ground fault), and duration 0.0833 s, approximately five cycles at 60 Hz.

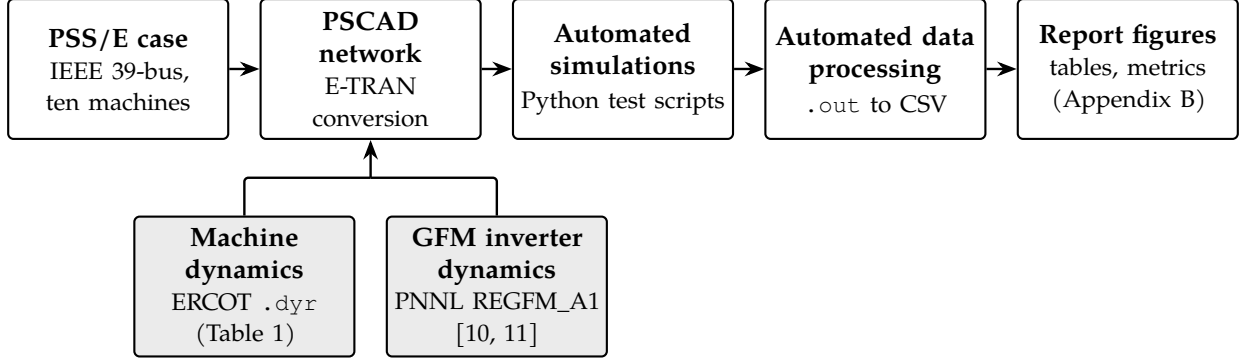


Figure 5. Testbed workflow. E-TRAN converts the IEEE 39-bus PSS/E case into a PSCAD EMT network; synchronous-machine dynamics from the ERCOT-supplied synchronous machine dynamics and grid-forming inverter dynamics from the PNNL REGFM_A1 model are inserted into that network; Python test scripts set up and run the simulations, and processing scripts convert the EMTDC output into tables, metrics, and the report’s figures. The interpretation of those results is not automated.

2.3 Measurement in Testbed Design

The instrumentation records per-bus instantaneous and root-mean-square (RMS) voltage and current, active and reactive powers, and frequency quantities. Frequency analysis includes the nadir, recovery, maximum deviation, and rate of change of frequency. We parse the PSCAD output files and normalize device quantities on their apparent-power bases for plotting. Physical-unit network plots and device-normalized plots retain their respective measurement locations and voltage levels.

Voltage and current interpretation depends on the selected bases and on whether the signal is instantaneous or RMS. We distinguish line-to-line voltage from line-to-neutral voltage when interpreting the three-phase quantities. Device-base normalization makes the controller settings readable but does not, by itself, make unequal ratings or operating points identical. Appendix B describes the channel-processing conventions.

Table 1 lists the governor and turbine settings used by the machine model. Appendix A gives the machine, excitation, governor, and stabilizer record in PSS/E format. These dynamic settings act alongside the network operating point to determine the machine response. The parameter units follow the corresponding normalized control signals [40, 41].

The machine model includes the rotor and field dynamics, an exciter, a turbine governor, and a power-system stabilizer. The GGOV1 settings specify permanent droop, the speed-error limits, valve motion, turbine response, and load and acceleration limits. The transducer time $T_{pelec} = 1.0$ s filters the electrical-power feedback selected by $Rselect = 1$; it is the supplied value, equal to the typical GGOV1 value tabulated in [40], and is distinct from the inverter’s 10 ms power-measurement filter and from any plotting filter. The proportional, integral, and derivative governor gains act on different time operations, so their units differ even though the input and output signals are normalized. Keeping these settings explicit connects the machine baseline to the damping comparison in Section 4.6.

2.4 GFM Inverter Models Evaluated in Simulation Testbed

We use the Pacific Northwest National Laboratory (PNNL) REGFM_A1 droop-controlled model in the system studies [10]. REGFM_B1, a virtual synchronous machine (VSM) variant, was also evaluated in model-quality and inertia tests [13]. PNNL provides these models for PSCAD and PSS/E [11]. The REGFM_A1 single-device tests include low-voltage ride-through, voltage and

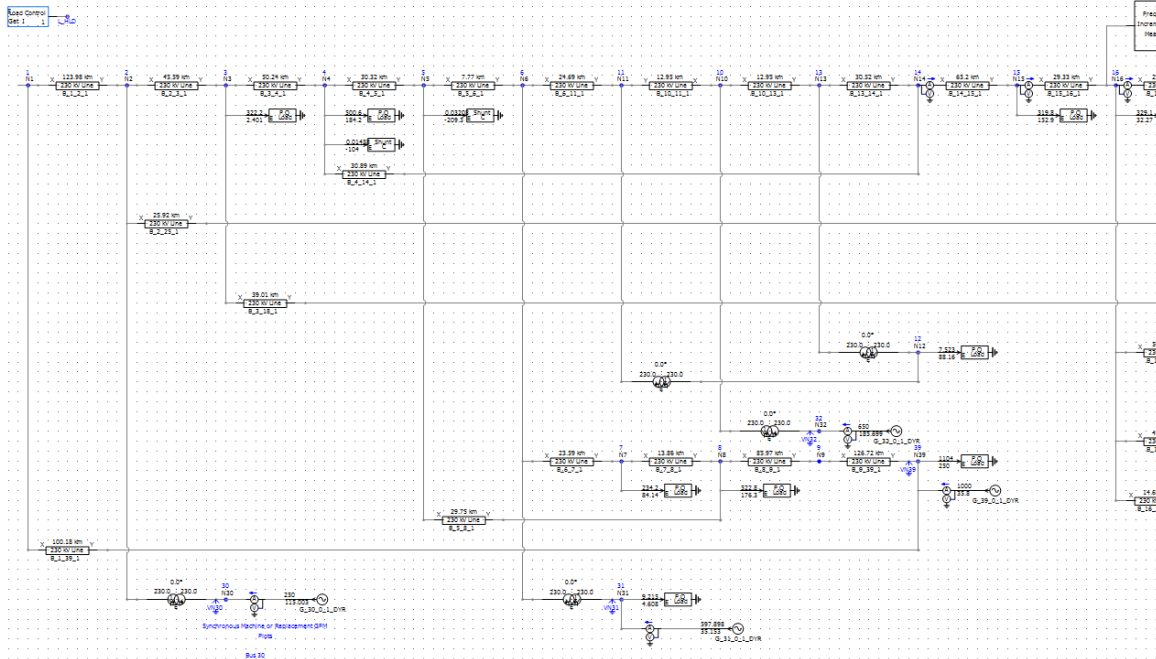


Figure 6. Single-line overview of the E-TRAN-converted IEEE 39-bus network on the PSCAD canvas: 230 kV lines with per-line length labels, transformers, and the ten generator subsystems.

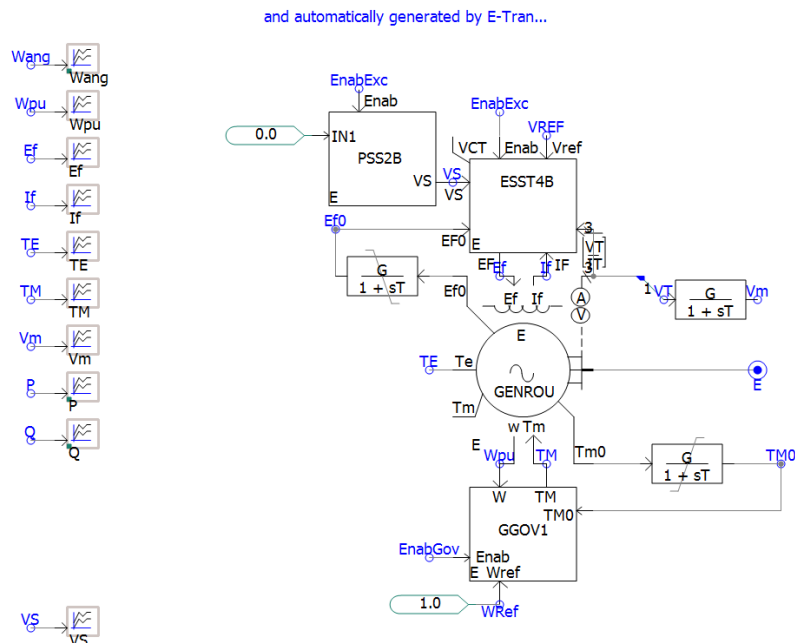


Figure 7. Machine subsystem internals as generated by E-TRAN from the .dyr record: the GENROU machine with its PSS2B stabilizer, ESST4B exciter, and GGOV1 governor blocks, plus the per-signal recorder taps.

Table 1. GGOV1 governor and turbine parameters for the synchronous-machine reference, in the order of the PSS/E GGOV1 record [41]; typical values and the block diagram are in [40].

Parameter	Value	Units	Definition
Rselect (ICON)	1	integer	Electrical-power droop feedback
Flag (ICON)	1	switch	Fuel-flow characteristic
R	0.05	p.u./p.u.	Permanent droop
T_{pelec}	1.0	s	Electrical-power transducer time
maxerr / minerr	± 0.055	p.u.	Governor speed-error clip (PID input)
K_{pgov}	10	p.u./p.u.	Proportional gain
K_{igov}	1	s^{-1}	Integral gain
K_{dgo}	0	s	Derivative gain
T_{dgo}	1	s	Derivative-filter time
v_{max}/v_{min}	1 / 0.16	p.u.	Valve-position limits
T_{act}	0.5	s	Actuator time constant
K_{turb}	1.5	p.u./p.u.	Turbine gain
W_{fnl}	0.2	p.u.	No-load fuel flow
T_b	0.1	s	Turbine lag time
T_c	0	s	Turbine lead time
T_{eng}	0	s	Engine transport lag
T_{load}	3	s	Load-limiter time constant
K_{pload}/K_{iload}	2 / 0.7	p.u./p.u. / s^{-1}	Load-limiter PI gains
L_{dref}	1	p.u.	Load-limiter reference
D_m	0	p.u./p.u.	Turbine speed sensitivity
R_{open}/R_{close}	0.1 / - 0.1	p.u./s	Valve motion limits
K_{imw}	0	s^{-1}	Power-controller reset gain; disabled
A_{set}	0.1	p.u./s	Acceleration setpoint
K_a/T_a	10 / 0.1	p.u. / s	Acceleration gain/filter time
T_{rate}	0	MW	Turbine-rating input
db	0.0002	p.u.	Speed-governor deadband
T_{sa}/T_{sb}	4 / 5	s / s	Temperature lead/lag times
R_{up}/R_{down}	99 / - 99	p.u./s	Load-limit ramp caps; inactive

phase-angle steps, and frequency disturbances; Figure 16 shows its fault response.

For each replacement, we use the sizing rule $S_{inv,base} = P_G/0.6$, where P_G is the replaced machine's MW dispatch and $S_{inv,base}$ is expressed in MVA. The selected inverter active-power reference is 0.6 per unit on that base, the PSCAD component's default (Figure 9); matching the replaced MW at that reference fixes the base, whereas keeping the machine rating as the base would have required changing the reference, which was not done in the first-year runs. The rule preserves the intended MW injection but does not necessarily preserve the original machine's MVA rating or headroom. The inverter connects behind the step-up transformer at the replaced generator location.

2.5 Model Quality Testing

Model quality testing (MQT) applies the perturbations of the ERCOT Dynamics Working Group procedure manual [15] to prospective inverter and machine models before interpreting their network responses. We used PMView, versions 2.4 and 3.5, as the playback and measurement harness [14]. PMView replaces the grid source with a programmable infinite-bus voltage source and applies the test profiles described by the Dynamics Working Group Procedure Manual, Revision 24 [15]. The tests characterize initialization, voltage regulation, ride-through, and response to angle changes.

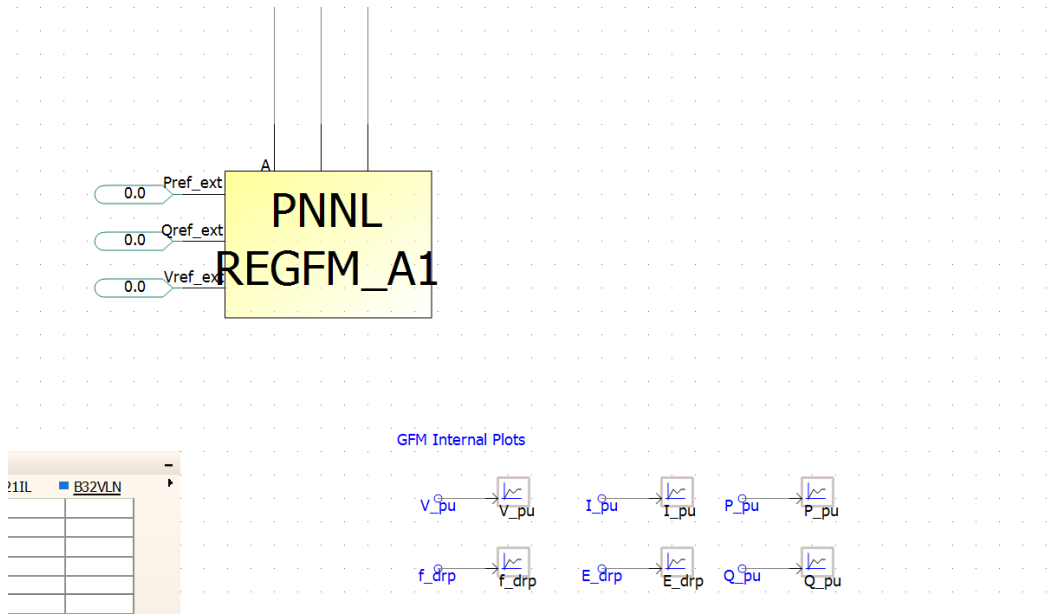


Figure 8. The PNNL REGFM_A1 component wired at a replaced machine bus, behind its 230/10 kV step-up transformer, with the external reference inputs and the internal-signal plot taps.

Tests 1–9 include flat start, legacy and IEEE 2800 low- and high-voltage ride-through, small voltage steps, and phase-angle jumps. PMView version 3.5 adds advanced grid-support tests, including the inertia estimation test discussed in Section 4.7. These device-level responses help distinguish controller behavior from interactions introduced by the network. The following models were included in the test program.

1. **PNNL REGFM_A1:** droop control with voltage regulation, current control, and current limiting [10].
2. **PNNL REGFM_B1:** a VSM variant evaluated in the model-quality and inertia estimation tests [13].
3. **National Laboratory of the Rockies (NLR) GFM model:** droop control with an integral voltage regulator [16].
4. **PSCAD library GFM model:** also included in the model-quality-testing program [17].

Four models were evaluated: REGFM_A1, REGFM_B1, the NLR GFM model, and the PSCAD library GFM model. REGFM_A1 was selected for the system studies as the configuration usable as shipped; the other models were not tuned further.

We applied the same nine-test battery to the synchronous machine used in the network studies. The machine completed these tests successfully and provides a conventional reference for the inverter comparisons. Figure 14 shows its response to the legacy low-voltage ride-through profile, corresponding to the REGFM_A1 example in Figure 13. The machine voltage-step result in Figure 15 uses a different step magnitude from the inverter comparison.

The machine example is the 650 MVA Gen-32 test case. Its small-step test raises voltage from 1.00 to 1.03 per unit at 3 s and produces a slower response than the selected REGFM_A1 comparison. Figure 16 additionally records the inverter-side LCL quantities during the REGFM_A1 ride-through test. Voltage, current, active power, reactive power, and droop frequency together describe how the device supplies the grid through the disturbance.

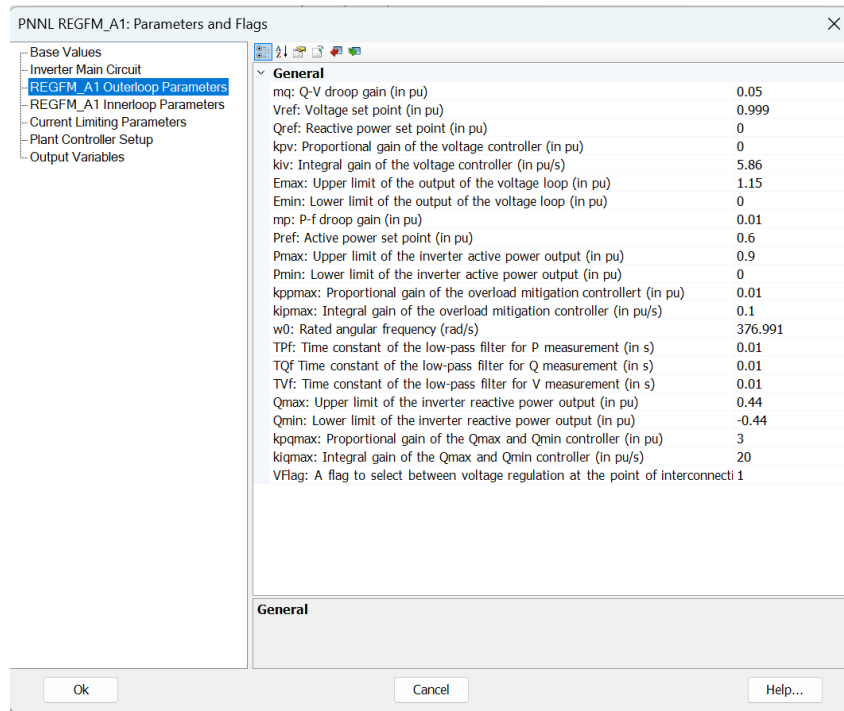


Figure 9. REGFM_A1 outer-loop parameter dialog as configured for the study: droop gains $m_p = 0.01$ and $m_q = 0.05$, filter time constants $T_{Pf} = T_{Qf} = 10$ ms, and voltage-loop ceiling $E_{\max} = 1.15$ p.u.

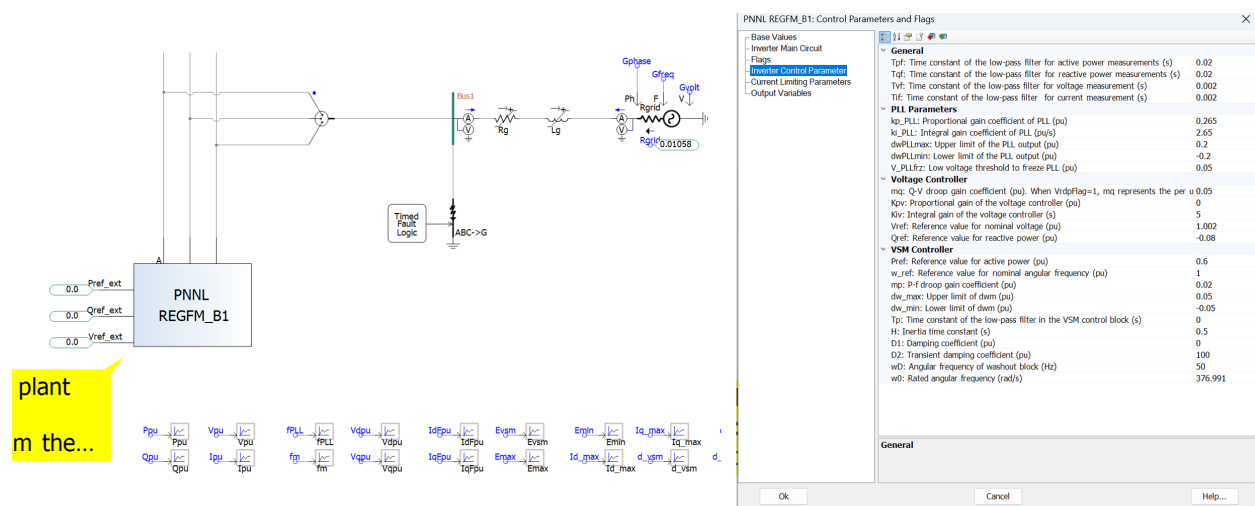


Figure 10. REGFM_B1 in its PMView-driven test harness (left) and its control-parameter dialog (right); the virtual-synchronous-machine block carries a configured equivalent inertia constant $H = 0.5$ s.

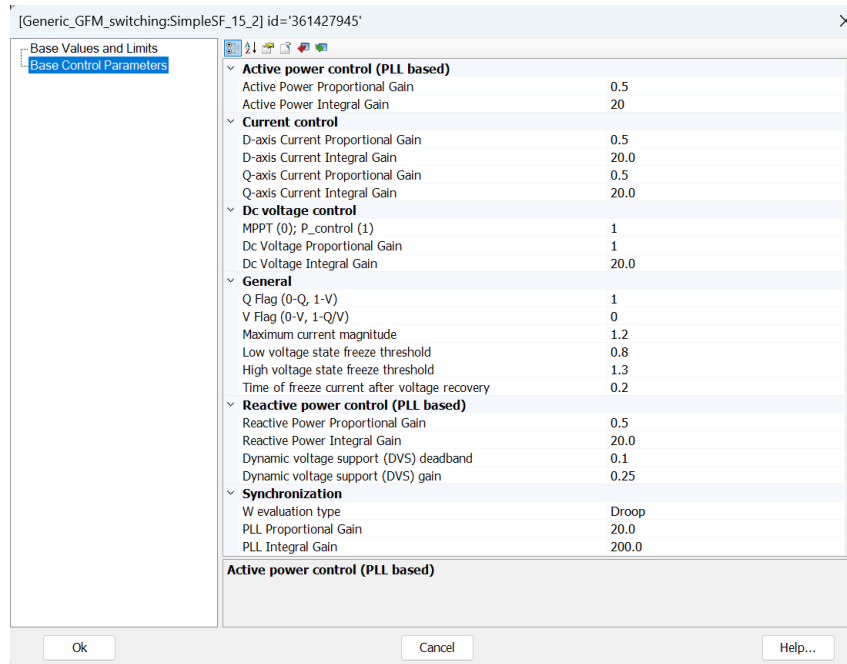


Figure 11. PSCAD library GFM model: control-parameter dialog of the aggregated inverter (droop synchronization, current and voltage loops, and limiter settings).

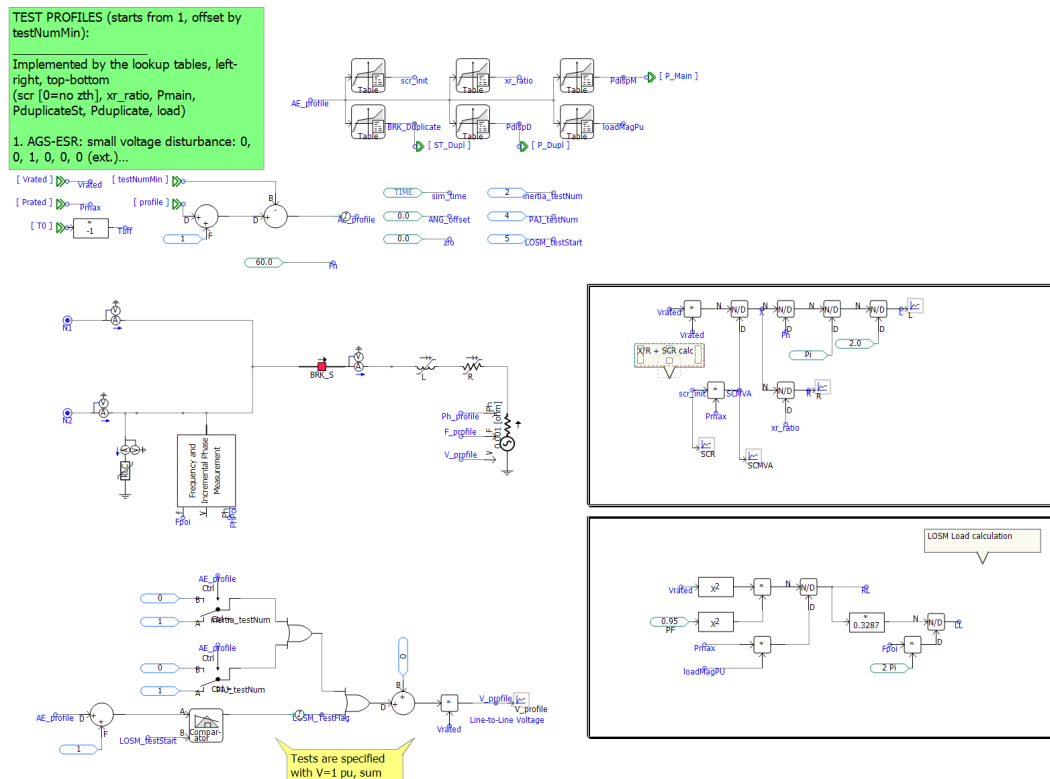


Figure 12. PMView test setup: the playback source and point-of-interconnection measurement wired around the device under test, with the per-test profile lookup tables that program the disturbance battery.

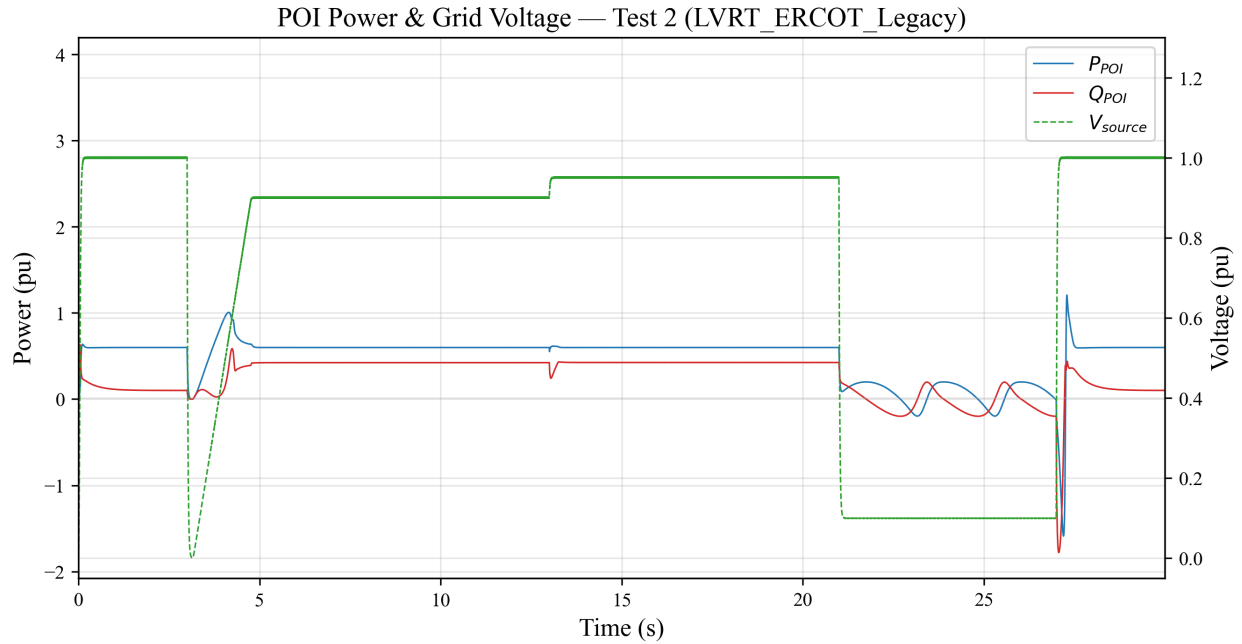


Figure 13. REGFM_A1 response to the legacy low-voltage ride-through profile in PMView.

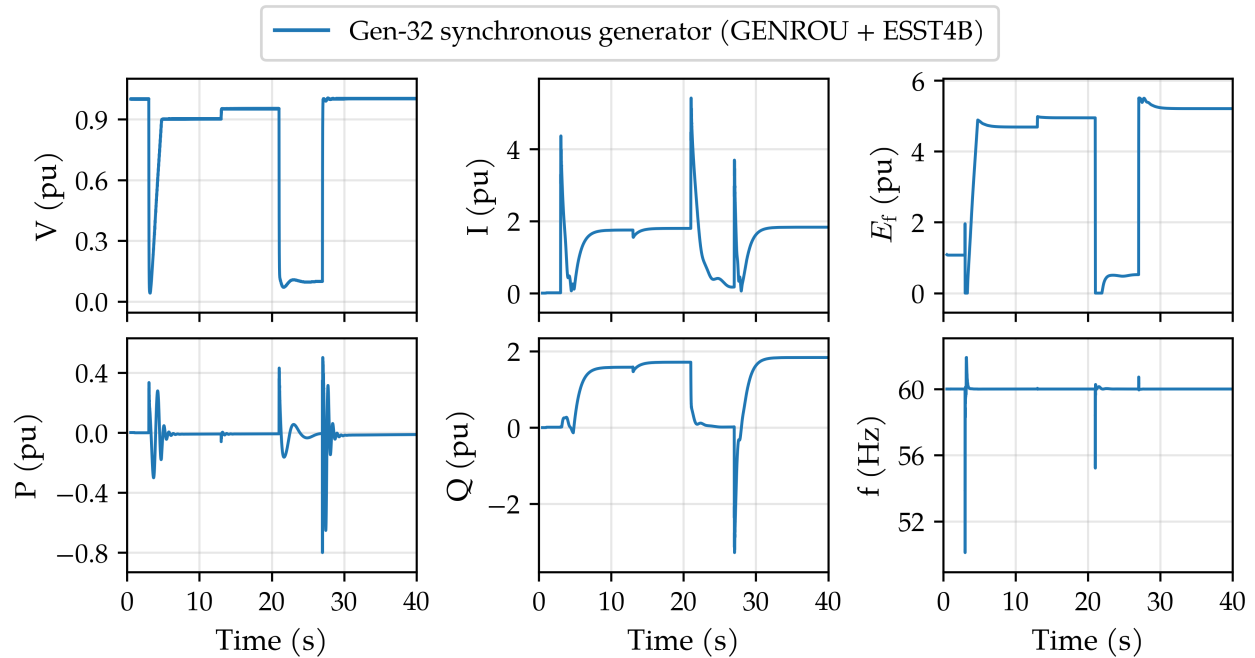


Figure 14. Synchronous-machine response under the legacy low-voltage ride-through profile for the Gen-32 test case.

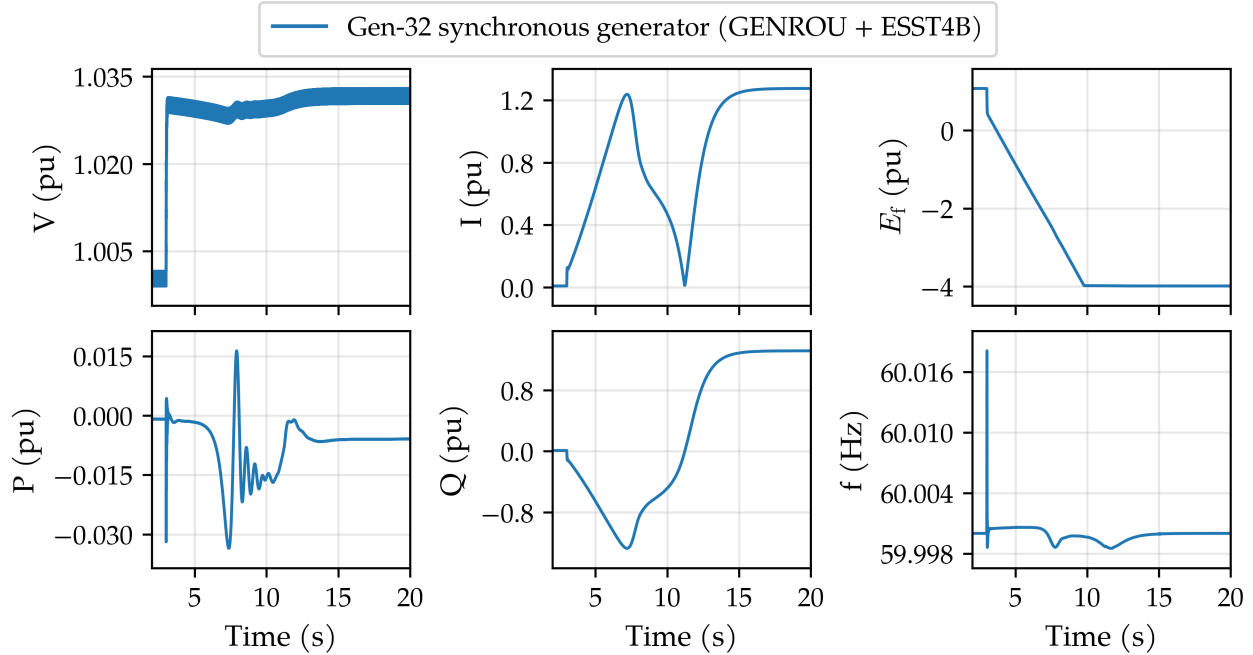


Figure 15. Gen-32 response to a source-voltage step from 1.00 to 1.03 per unit at 3 s.

2.6 Study Design

We first applied a three-phase-to-ground fault at bus 14 to the all-machine system. The fault resistance is 0.01Ω , a low-impedance approximation to a bolted fault. Initial machine studies considered both restoration of the faulted connection and clearing with the connection left open. The reference case used for the replacement progression has a five-cycle fault without reclosing.

We selected a short reference disturbance to represent a practical protection timescale. The 0.0833 s duration corresponds to approximately five cycles at 60 Hz and is applied from 3.0 s. After establishing the machine baseline, we replaced machines one at a time and reapplied the same reference fault. Extended-duration tests discussed later explore transient stability limits rather than recommended protection operating times. Table 2 lists the simulated cases. Network faults were applied at bus 14 and bus 10 only; other fault locations were not tested. All faults in the replacement progression and the current-limit studies are three-phase-to-ground; single-line-to-ground and double-line-to-ground checks at bus 14 are reported in Section 3.5. No overcurrent trip is modeled in any case, and instantaneous failures of the semiconductor switches, which occur on a much faster timescale than the control limiter, are not represented.

2.7 PSCAD Automation

We configured and ran the one-for-one replacement cases manually. For subsequent model-quality studies, we used PSCAD's `mhi.pscad` Python interface to launch PSCAD, load the case, modify parameters, and export results. Component identifiers locate the fault application time, duration, resistance, solution step, and run length. Each run writes a separate directory named for the varied parameter and stores the exported channels with its fault configuration and device parameters.

The automation sets parameters before launching each simulation and collects the output channels after completion. A written configuration accompanies each run so that a parameter sweep can be interpreted as a sequence of distinct test cases. We used this workflow for repeated device

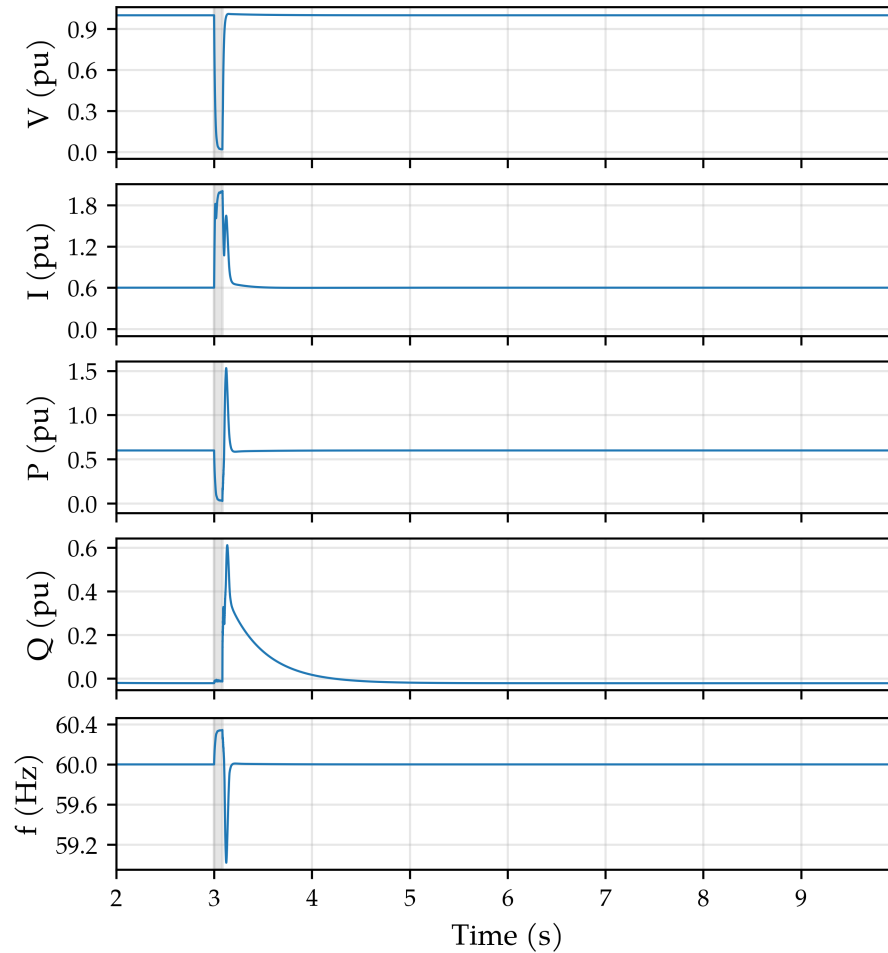


Figure 16. Single REGFM_A1 fault response with a 2.0 per-unit transient-current limit: normalized voltage, current, active and reactive powers, and frequency in Hz.

tests while retaining the manually configured replacement cases as the network-study starting point.

2.8 Data Processing

At the end of a run, PSCAD writes numbered `.out` files containing whitespace-delimited numerical channels. The first column is time, followed by up to ten recorded channels, so a large case produces multiple files with repeated time columns. A companion `.inf` file supplies the global channel indices, descriptions, groups, limits, and units.

The processing pipeline reads the `.inf` file and reconstructs the mapping from channel indices to the numbered output files. It then combines the channels into a common time-indexed table. Descriptions are not necessarily unique: different phases or measurement locations can share the same name. The pipeline distinguishes these descriptions before exporting CSV files for analysis and plotting.

The plotting scripts read the resulting tables to draw the simulation responses. Screenshots, circuit diagrams, and analytical illustrations complement those numerical plots. Appendix B describes the file formats and plotting conventions. The workflow allows the same processing approach to serve both individual-device and network studies.

Table 2. Simulated fault cases. Fault resistance 0.01 Ω ; fault applied at 3.0 s.

Study	Configurations	Fault bus	Duration; clearing	GFM $I_{\max F}$
Replacement progression (Section 3)	0–10 GFM units	14	5 cycles; lines to bus 14 opened, no reclosing	2.0
Current-limit comparison	0 and 10 GFM units	10	5 cycles; lines to bus 10 opened, no reclosing	1.5
Current-limit sensitivity (Section 3)	10 GFM units	14	5 cycles; lines to bus 14 opened, no reclosing	2.0; 1.5; 1.2
Unbalanced-fault check (Section 3.5)	0 and 10 GFM units	14	5 cycles, single-line-to-ground and double-line-to-ground; lines to bus 14 opened, no reclosing	2.0
Single-device comparison (Section 4)	one machine; one inverter; infinite bus, SCR 10	point of interconnection	5 cycles; self-clearing	2.0
Clearing-time limits (Section 5.1.1)	ten machines; ten GFM units; one machine; one inverter	14; point of interconnection	duration swept; self-clearing, network intact	2.0; 2.0–15 (single inverter)

The simulation records behind the figures, the processing and plotting scripts, the figure data, and the PSCAD test systems are in the project’s public repository [12]. The PNNL REGFM_A1 model used throughout, together with the REGFM_B1, REGFM_C1, and REPCGFM_C1 models of the same family, is publicly available in PNNL’s release of the WECC grid-forming inverter models [11].

3 Simulation Testbed Results

The results compare fault recovery as machines are replaced by GFM inverters and examine the role of current limiting. We used PSCAD 5.0.2 with GFortran 4.6 and the E-TRAN library for the converted IEEE 39-bus network [36]. The REGFM_A1 model uses the PNNL library `PNNL_REGFM_A1_gf46.lib` [37]. The fault-sweep cases use a 50 μ s solution time step.

The reference fault begins at 3.0 s and lasts five cycles. Figure 17 shows the first two steps of the replacement progression, with one and two machines replaced. Figure 18 shows the all-GFM arrangement. The bus-14 reference study uses the 2.0 per-unit transient-current setting, while the separate bus-10 current-limit comparison uses 1.5 per unit. Clearing opens the breakers of the lines connected to bus 14 and isolates that bus: after clearing its active power, reactive power, and current are zero, and its plotted 30 kV is the residual voltage of the de-energized bus rather than a network-recovery quantity.

3.1 Iterative Synchronous Machine Replacement

The eleven replacement configurations show recovery of the connected network under the reference disturbance. Figures 19–24 compare the all-GFM response with the machine baseline and selected

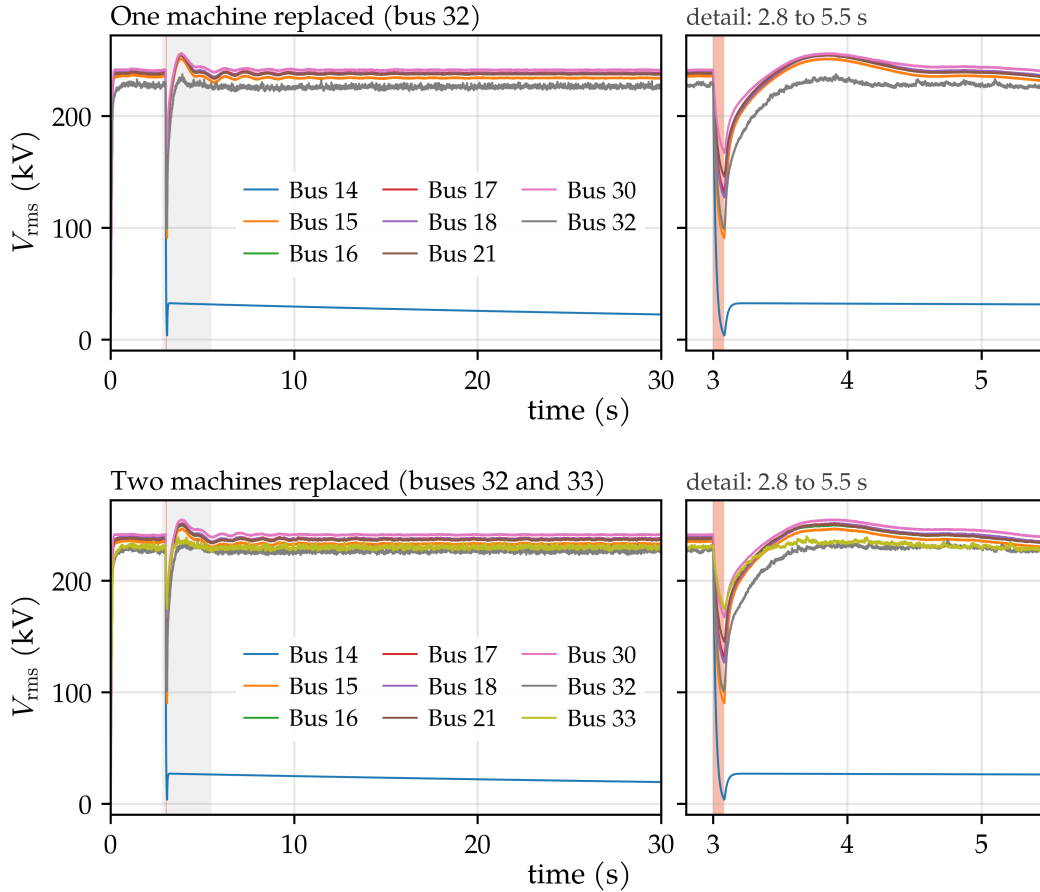


Figure 17. Bus-voltage responses with one replaced machine (top) and two replaced machines (bottom) under the bus-14 reference fault. Bus 14 is isolated by clearing; its trace is the residual voltage of the de-energized bus.

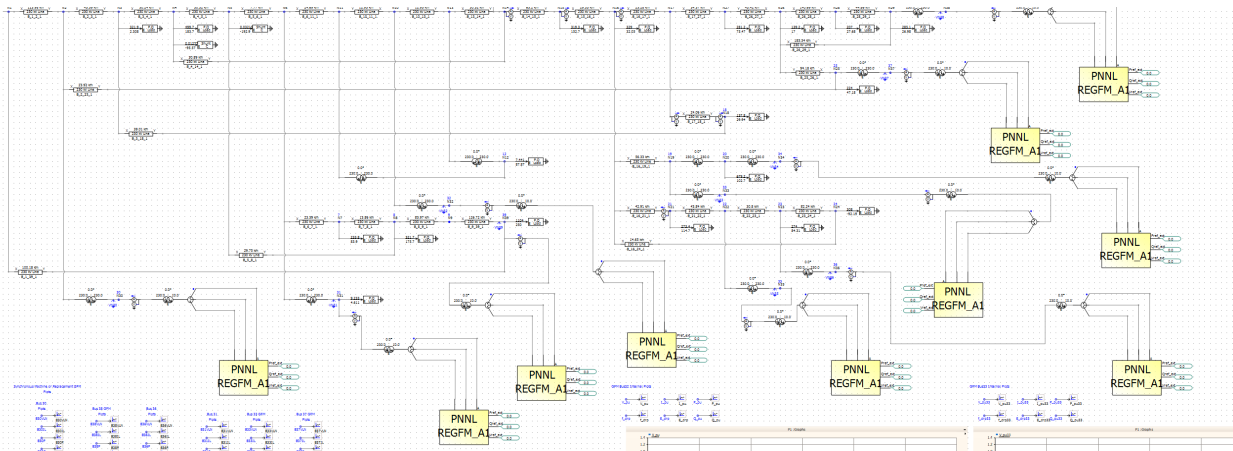


Figure 18. The fully inverter-based 39-bus network on the PSCAD canvas: all ten machine subsystems replaced by PNNL REGFM_A1 units (yellow blocks), each behind its step-up transformer, with the per-bus signal taps and system plot panels.

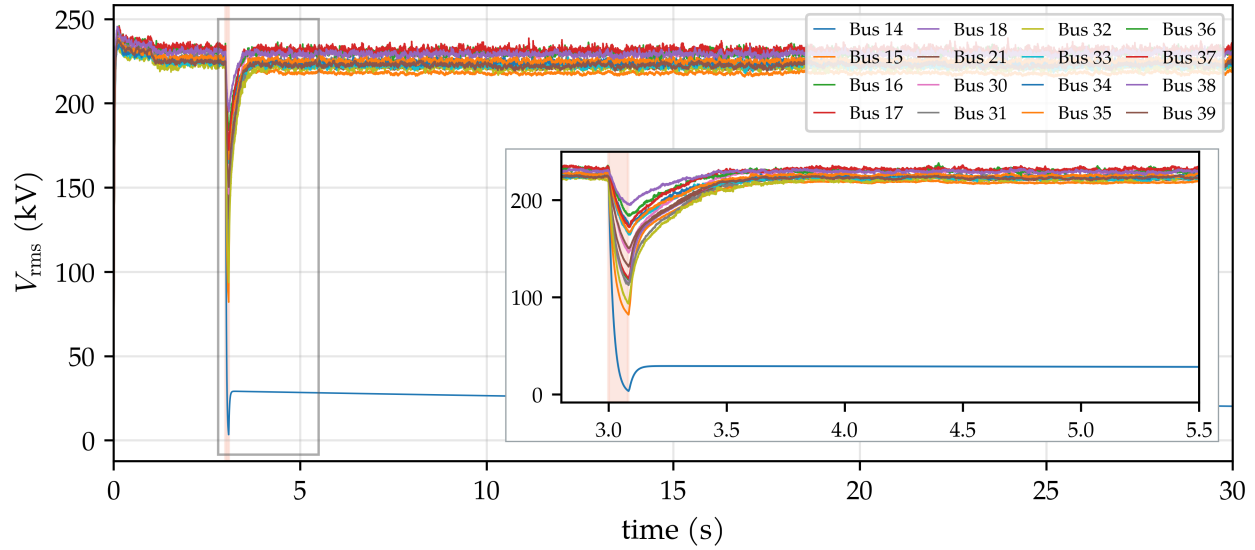


Figure 19. All-GFM bus-voltage responses to the bus-14 reference fault. RMS voltages are shown in kV for the monitored network locations.

intermediate configurations. Breaker opening removes the faulted connection, and reclosing is disabled in the reference progression. Reclosing can be added through the testbed's breaker logic for other study conditions.

The selected GFM responses return promptly toward their post-fault operating points. The machine response contains longer oscillations, while the voltage and power trajectories also vary by location. Opening a line can change the post-fault operating point, so recovery need not return every quantity to its pre-fault value. The progression demonstrates mixed-resource studies under a common disturbance without assuming that every placement or dispatch behaves identically.

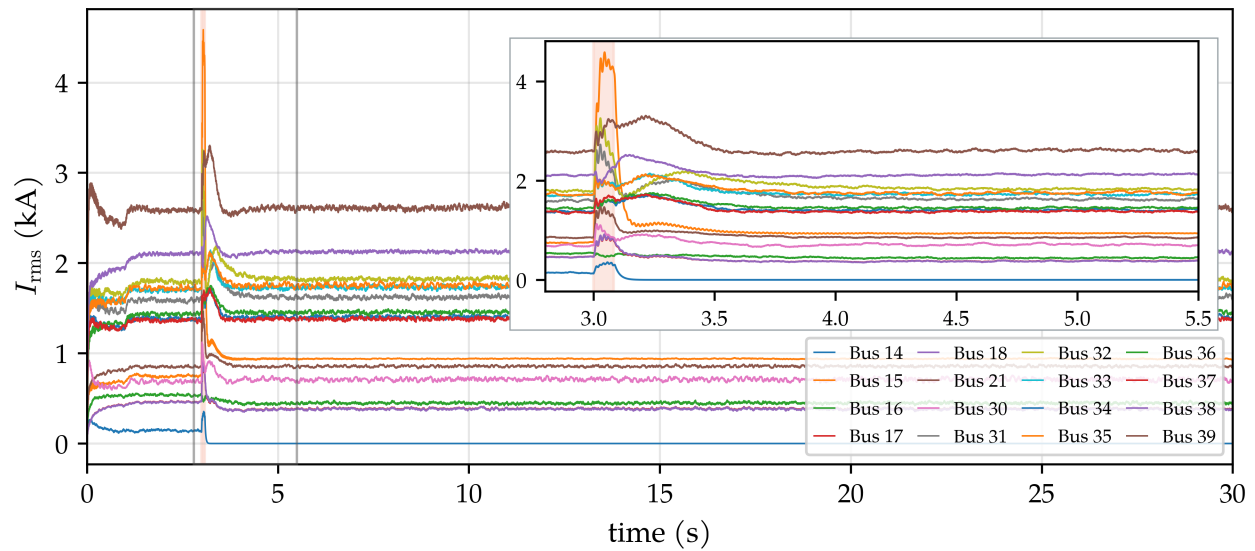


Figure 20. All-GFM RMS current responses to the bus-14 reference fault, in kA, measured on the 230 kV side of each unit's step-up transformer. The inverter transient-current setting is $I_{\max F} = 2.0$ per unit on each device base.

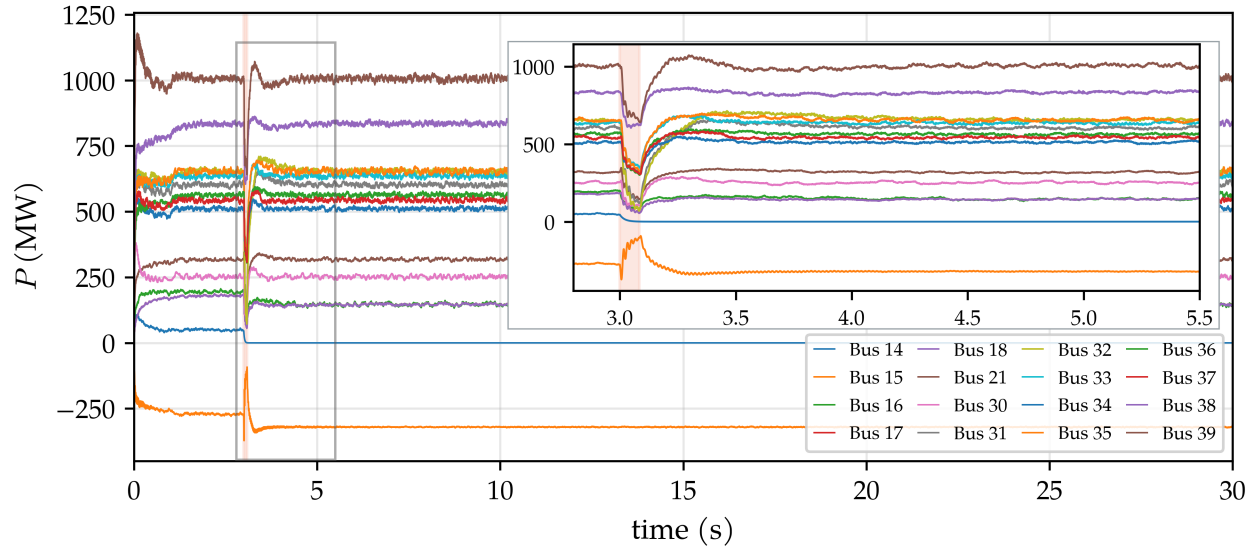


Figure 21. All-GFM active-power responses to the bus-14 reference fault, in MW, measured on the 230 kV side of each unit's step-up transformer. Signed powers follow the measurement-channel direction.

3.2 Defining GFM Current-Limiting Behavior

The bus-10 study uses a GFM transient-current setting of 1.5 per unit. Eight units remain at or below that setting, bus 31 reaches the limit, and bus 32 reaches approximately 1.70 per unit, 13% above the setting, within one cycle of fault inception in the inverter's internal current-magnitude signal, the quantity the limiter compares with $I_{\max} F$. Its terminal RMS current peaks at 1.24 per unit, and its instantaneous phase currents exceed 2.0 per unit of rated peak for at most 3 ms. Figure 25 follows these currents through the full 30 s record. After the transient, the connected units remain below 1.1 per unit and settle near 0.7–0.8 per unit.

The bus-32 current falls to zero after clearing, while the other nine units continue carrying

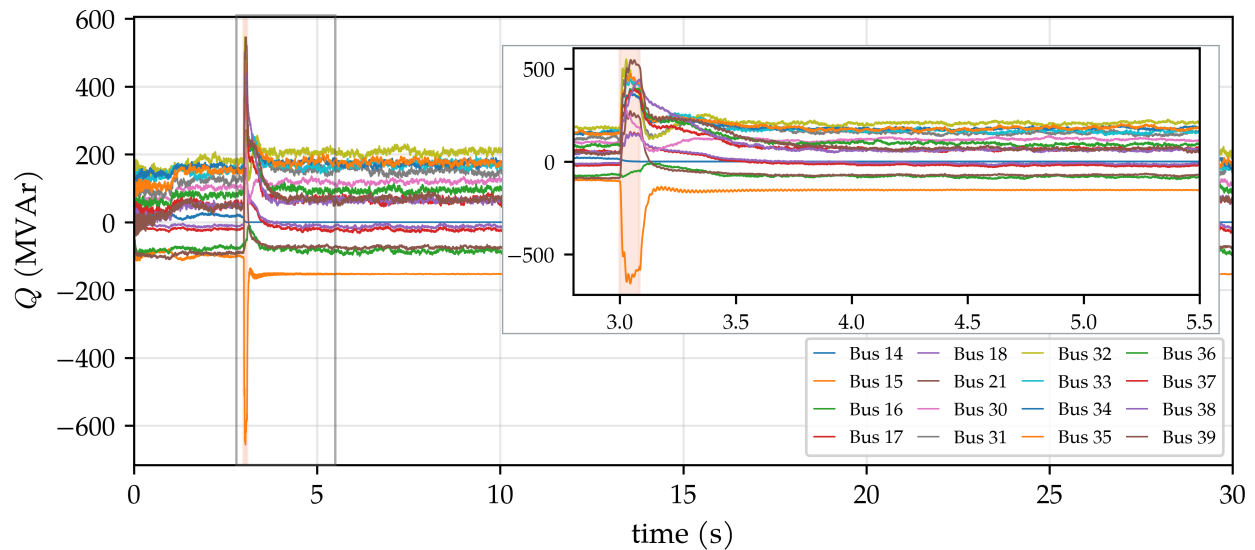


Figure 22. All-GFM reactive-power responses to the bus-14 reference fault, in MVar, measured on the 230 kV side of each unit's step-up transformer.

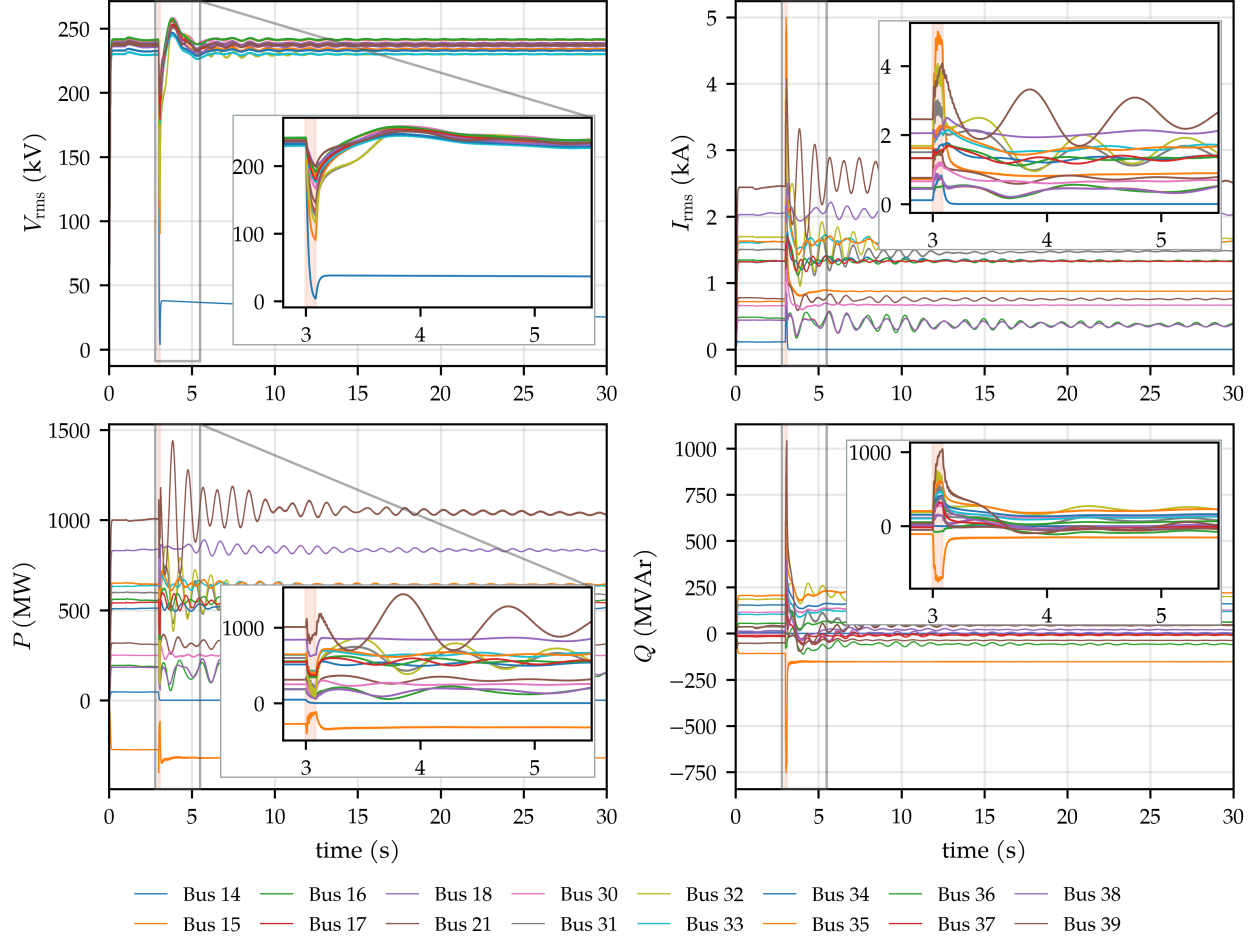


Figure 23. All-machine response to the bus-14 reference fault: RMS voltage and current, active power, and reactive power at the 230 kV generator buses, on the transmission side of the step-up transformers. The generator-bus active powers oscillate together at about 1 Hz, the inter-machine swing mode, and decay over tens of seconds. The legend lists the monitored buses; bus 14 is isolated by clearing.

current. The case associates this loss of output with opening the bus-10 network connections without reclosing. Continued network recovery therefore does not mean that the zero-current resource continues exporting power. This distinction is important when interpreting the fleet response after a clearing action.

Figure 26 compares the bus-14 reference fault on the all-GFM fleet at $I_{\max F} = 2.0, 1.5,$ and 1.2 per unit with all other settings unchanged. The fault is electrically remote from the units: at 2.0 per unit no unit reaches its limit, with peaks of 1.53, 1.44, and 1.38 per unit at buses 32, 30, and 31 and the other seven below 0.92 per unit. At 1.5 per unit only the bus-32 unit touches the limit, for 1.4 ms. At 1.2 per unit the three nearest units exceed the limit in the internal current-magnitude signal, with peaks of 1.44, 1.38, and 1.31 per unit at buses 32, 30, and 31; the limiter returns the current to the limit within half a cycle at buses 30 and 31 and within 1.7 cycles at bus 32, whose current makes a second, smaller excursion, so the total time above the limit is 0.4 to 0.8 cycle. Instantaneous failures of the semiconductor switches, which occur on a much faster timescale than the limiter's correction, are not modeled: the units stay connected and the control limiter alone bounds the current. In all three cases every unit rides through: post-fault active power returns to within 0.5% of its pre-fault value and the frequency settles at 60.01 Hz.

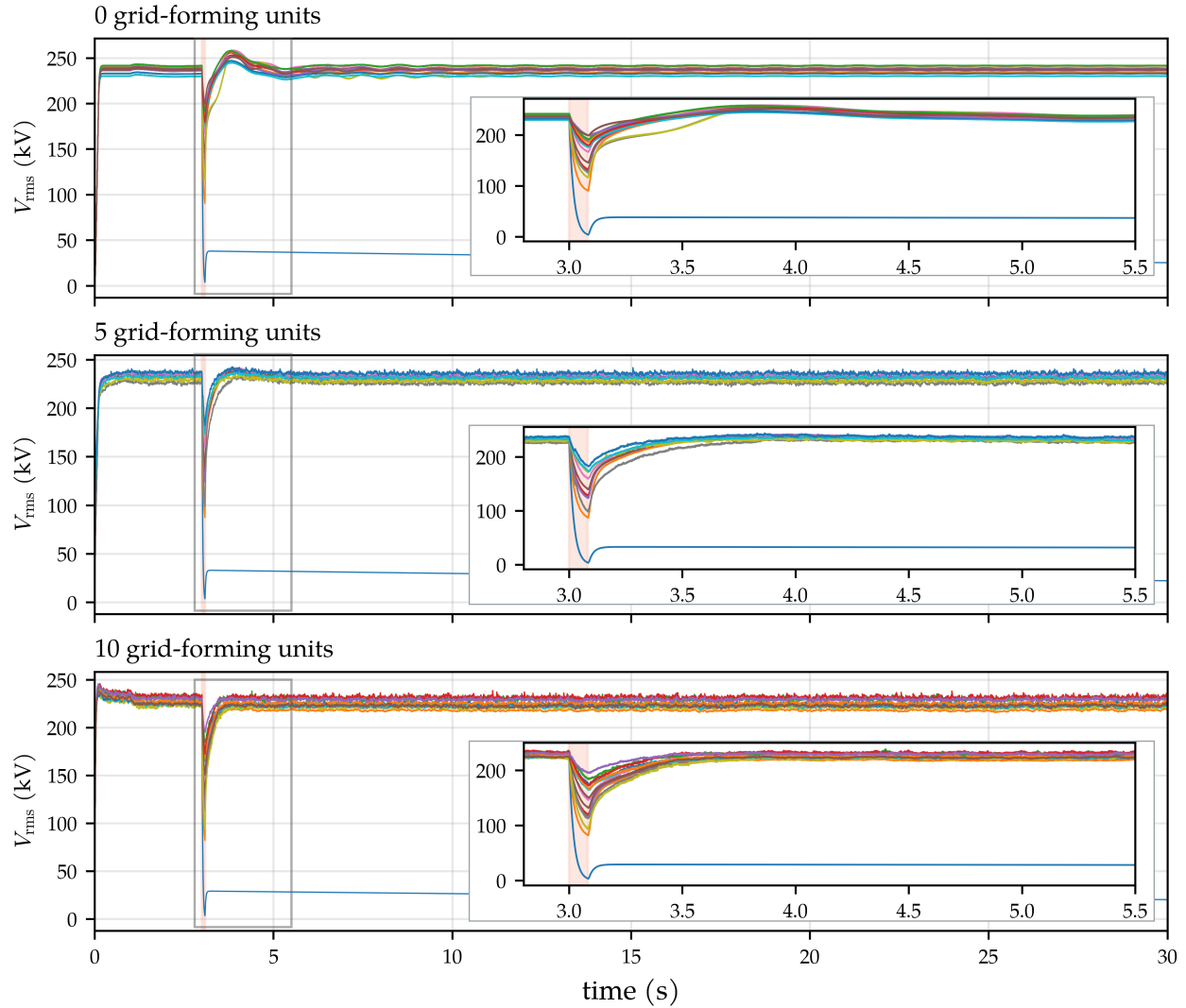


Figure 24. Bus-voltage responses with zero, five, and ten GFM units under the reference fault. The GFM transient-current setting is 2.0 per unit; configurations are identified by replacement count.

Figure 27 compares the device-base peak currents, including a machine peak of 3.72 per unit at the corresponding bus. GFM current is shaped by the modeled current limit, whereas machine fault current depends strongly on electromagnetic reactances and the network. Figure 28 shows the bus-39 terminal quantities, where the GFM case recovers more quickly and the machine current is substantially larger. Different ratings, dispatch, and measurement locations remain relevant even when both traces are expressed per unit. In the GFM runs, the active and reactive power traces immediately after clearing contain a 2–4 Hz settling component and a 60 Hz component below 10% of the deviation, the latter from the power calculation during the transient; the machine traces oscillate at the 0.85–1.0 Hz inter-machine mode.

3.3 All-Synchronous Machine Fault Ride Through

The all-machine system rides through the same reference fault as the all-GFM system. Its generator-bus active powers oscillate together at about 1 Hz, the inter-machine swing mode, and the oscillation persists for tens of seconds; the units nearest the fault, buses 31 and 32, settle faster. The stronger

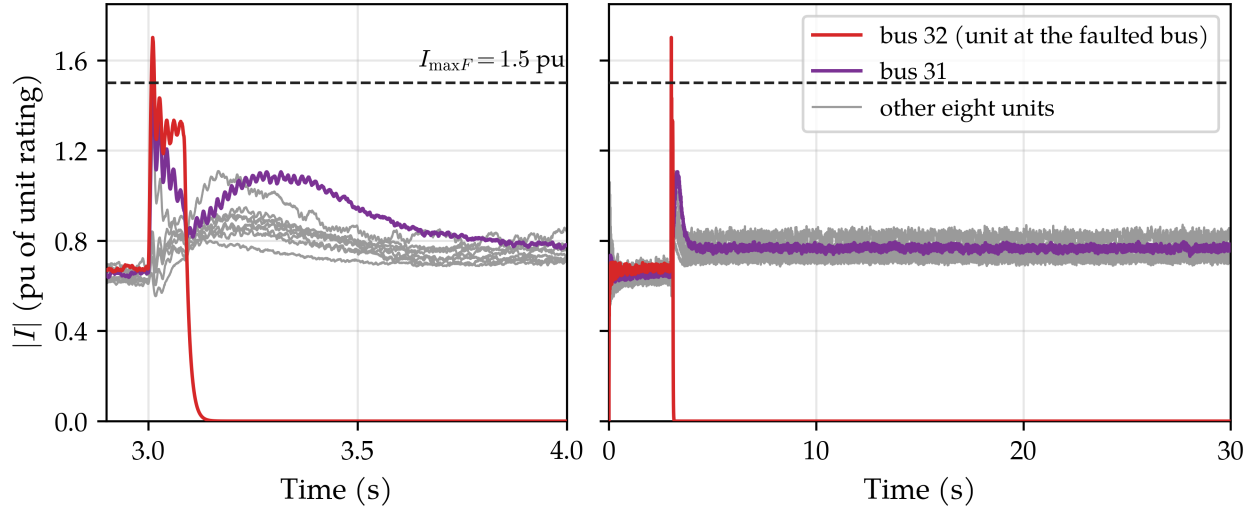


Figure 25. Bus-10 current-limit study with a five-cycle fault and no reclosing. Each trace is the inverter's own current magnitude at its terminals, on the 10 kV side of its step-up transformer, in per unit of the unit rating. With a 1.5 per-unit setting, bus 32 reaches approximately 1.70 per unit before falling to zero; nine units continue carrying current.

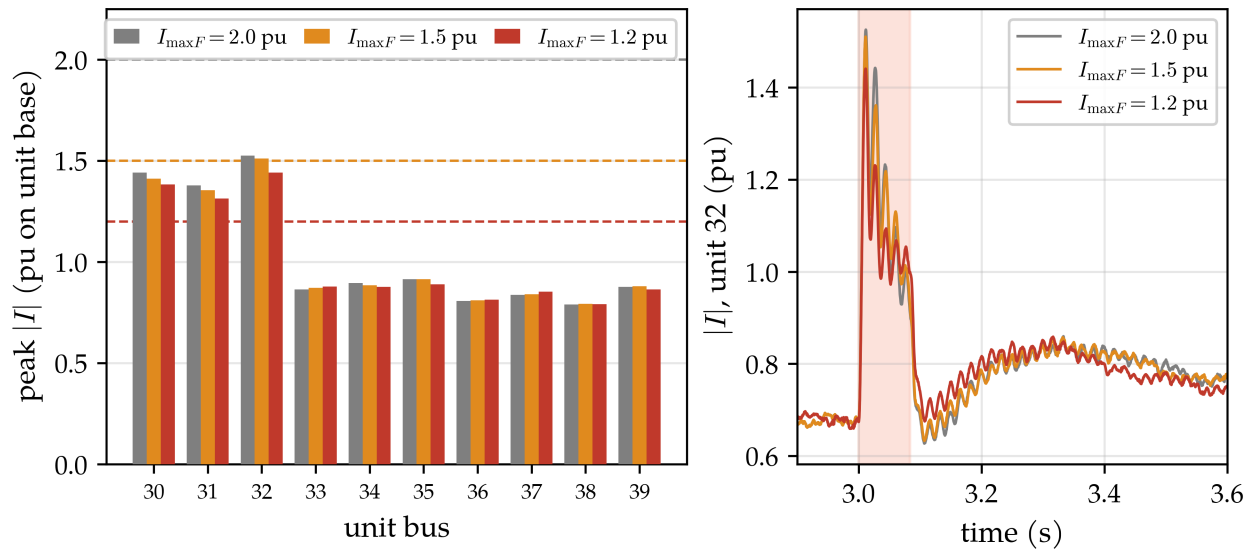


Figure 26. Bus-14 reference fault on the all-GFM fleet at three transient current limits, 10 s records with identical dispatch, fault, and clearing. Left: peak of each unit's internal current magnitude (the inverter's own measurement at its terminals, 10 kV side of the step-up transformer) on its own base, with the three limit lines. Right: the same signal for the bus-32 unit through the fault. Instantaneous failures of the semiconductor switches are not modeled.

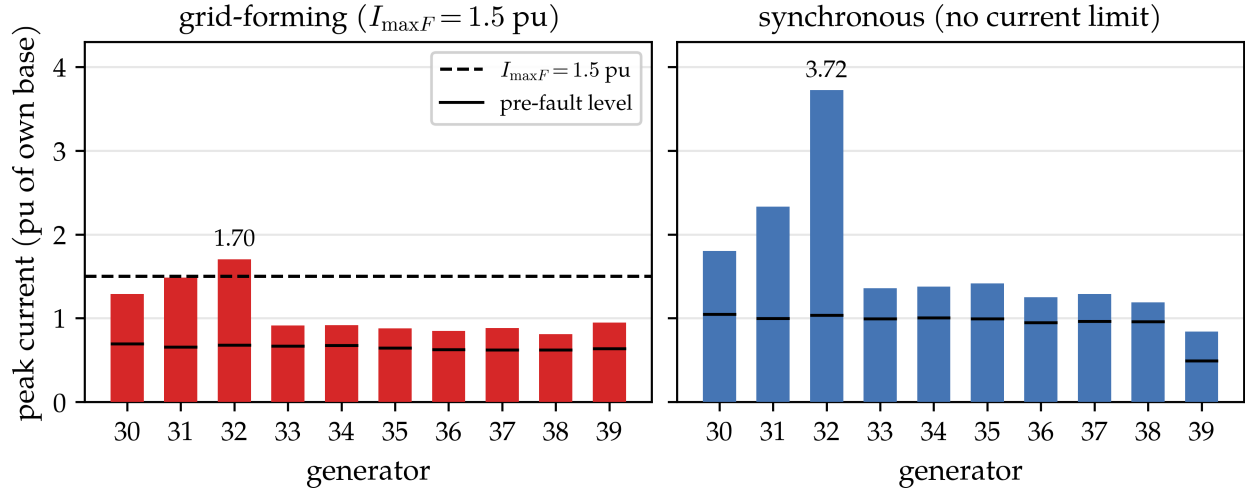


Figure 27. Peak currents for the bus-10 disturbance, normalized on individual device bases. For the GFM units the quantity is the inverter’s own current magnitude at its terminals, on the 10 kV side of the step-up transformer; for the machines it is the RMS current on the 230 kV side of the step-up transformer. The GFM transient-current setting is 1.5 per unit.

damping associated with the selected GFM controls helps interpret the contrast, as developed in Section 4. Network conditions, voltage controls, and limiting also affect the complete transient response.

3.4 Grid-Following Inverter Fault Ride Through

We also evaluated a PNNL-based GFL implementation in the testbed. The surrounding grid recovers within approximately one second, but the local voltage remains depressed over the plotted interval in Figures 29 and 30. The filtered voltage remains below the dip threshold and the active and reactive current commands fall to zero. These signals point to a control-state response rather than demonstrating semiconductor breakdown.

We obtained post-fault operation in a modified implementation that replaced the current-source interface with a voltage-source representation. That modification changes the model under test and does not establish a general remedy for GFL control. We did not pursue the GFL study further in the first-year work. The original and modified outcomes therefore remain distinct from the main GFM replacement results.

3.5 Unbalanced Faults at Bus 14

Single-line-to-ground (phase A) and double-line-to-ground (phases A and B) faults were applied at bus 14 with the reference timing and clearing on the all-machine and all-GFM fleets, 10 s records (Table 2). Figure 31 shows the bus-16 RMS voltage through the fault and the peak RMS current of each unit on its own base. For the machines the peak unit currents are 1.51 and 2.22 per unit at bus 32 for the single-line and double-line faults, with instantaneous phase-current peaks of 2.14 and 4.12 per unit of rated peak, and the lowest monitored RMS voltage during the fault is 0.82 and 0.60 per unit at bus 15. For the GFM units the peak RMS currents are 0.85 and 1.08 per unit at bus 32, the internal current-magnitude signal stays below the 2.0 per-unit limit (peaks 0.95 and 1.40 per unit), and the lowest monitored RMS voltage is 0.81 and 0.57 per unit. In all four cases every unit rides through: post-fault active power returns to within 3.8% (machines) and 0.6% (GFM units) of its pre-fault value on every unit, and the frequency settles at 59.84–59.90 Hz for the machines and

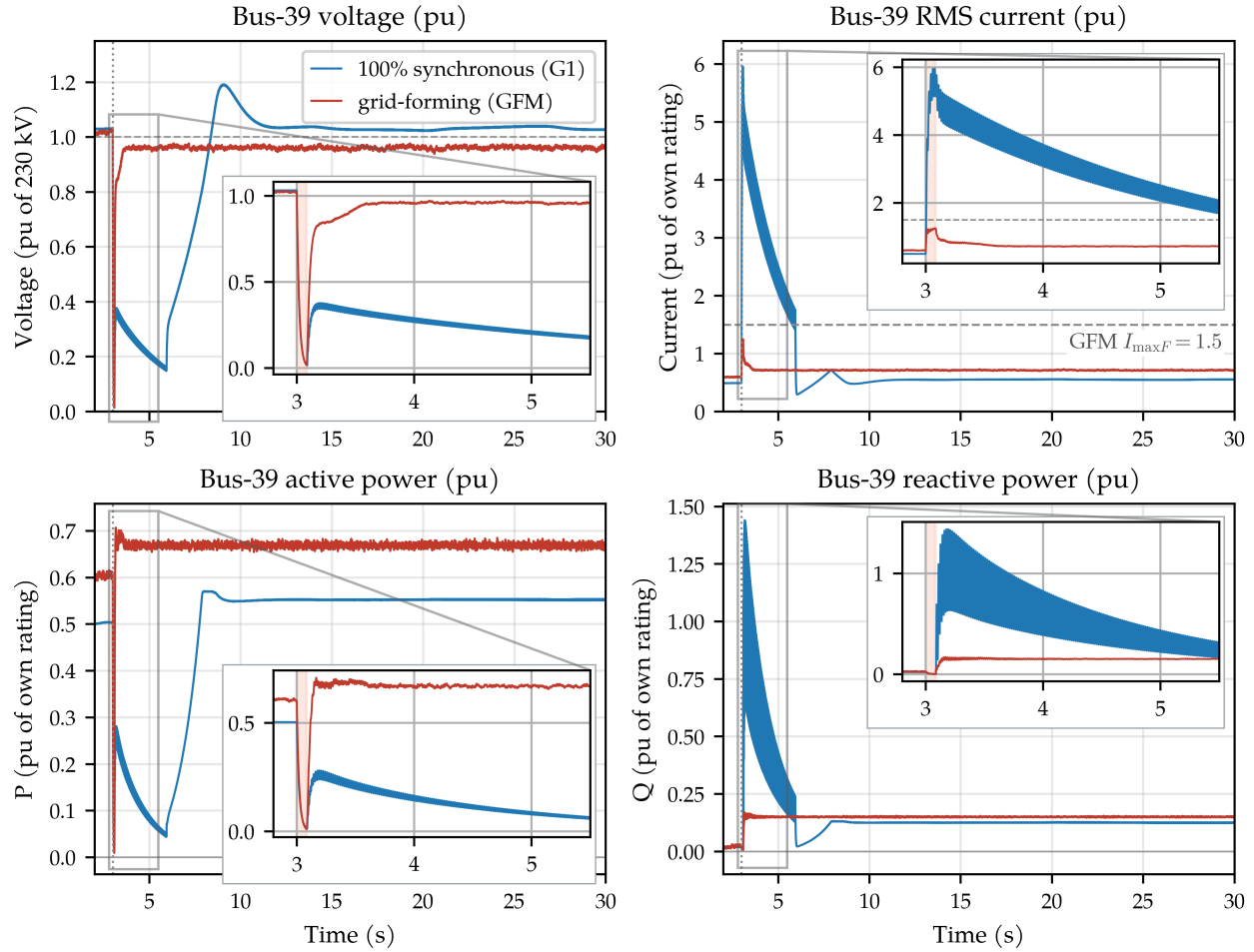


Figure 28. Bus-39 responses in the bus-10 study, measured on the 230 kV side of the unit's step-up transformer: all-machine (blue) and all-GFM (red), each on its own device base. The GFM case shows smaller fault current and faster settling; the two cases have different pre-fault power injections.

60.00 Hz for the GFM fleet.

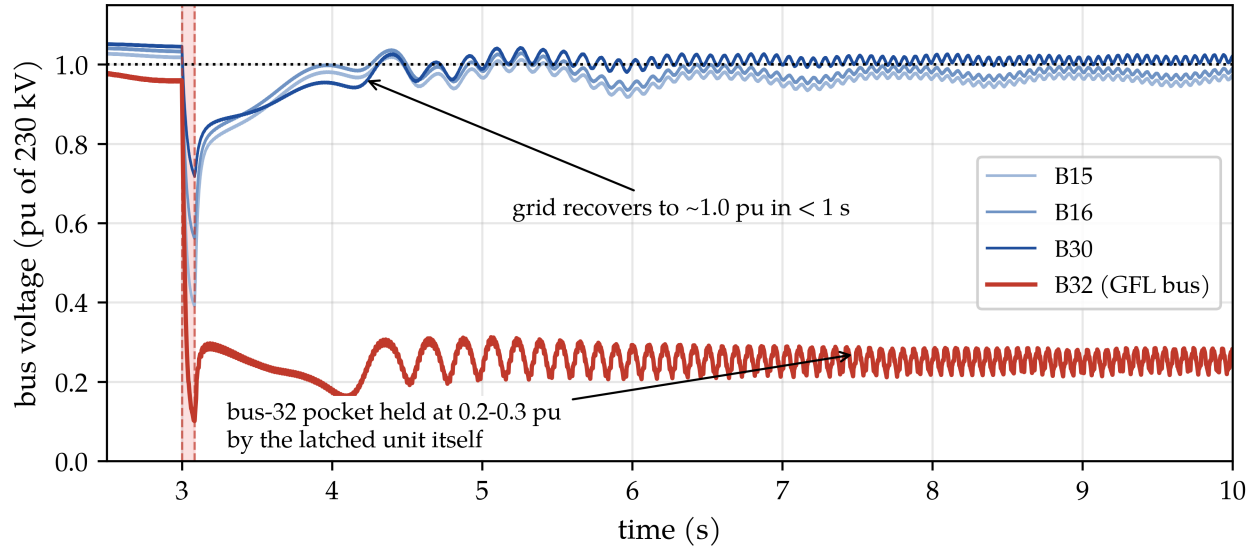


Figure 29. Original GFL implementation at bus 32: surrounding bus voltages recover while the local voltage remains depressed over the plotted interval.

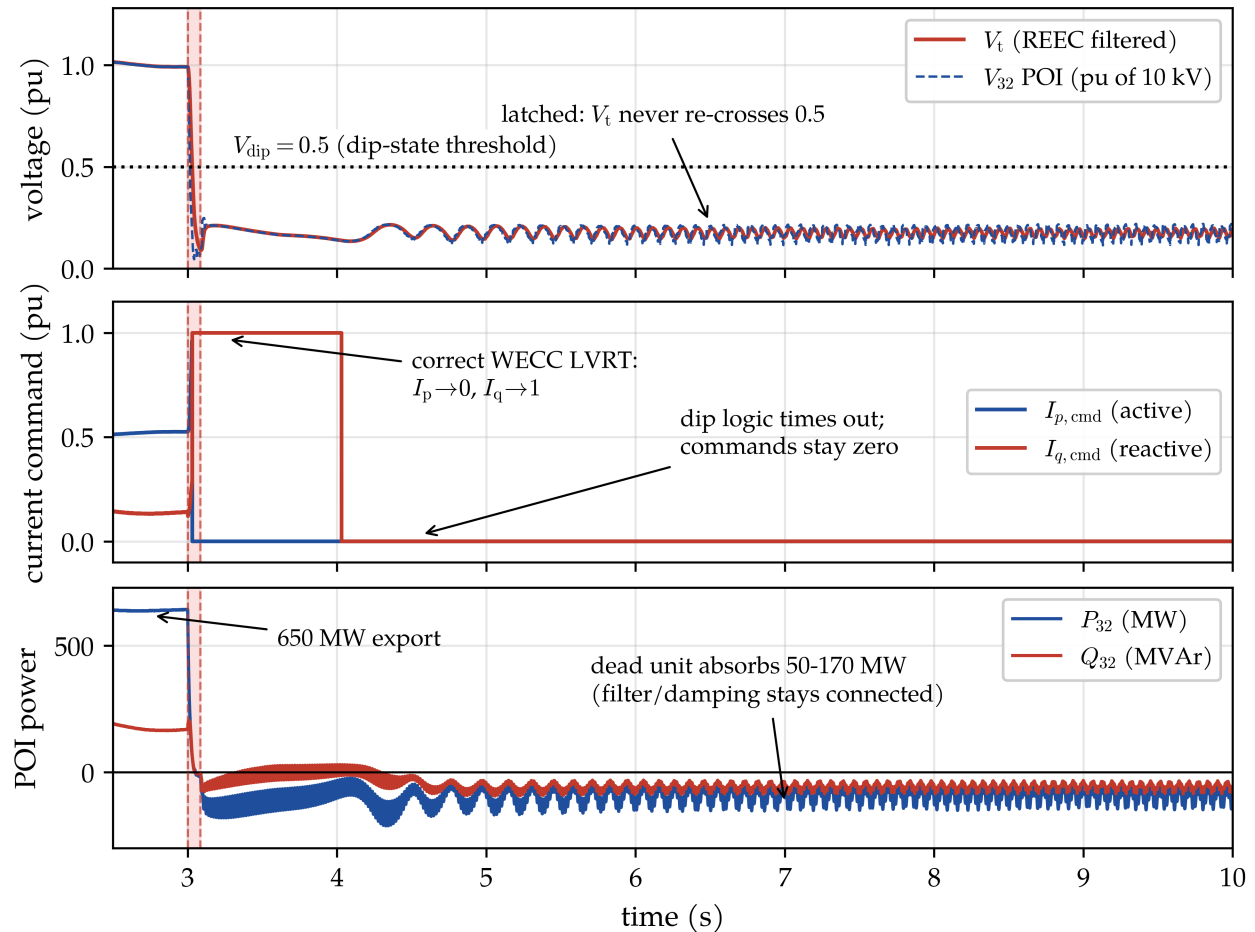


Figure 30. Voltage-dip and current-command signals in the original GFL implementation. Filtered voltage remains below the dip threshold, followed by zero active and reactive current commands.

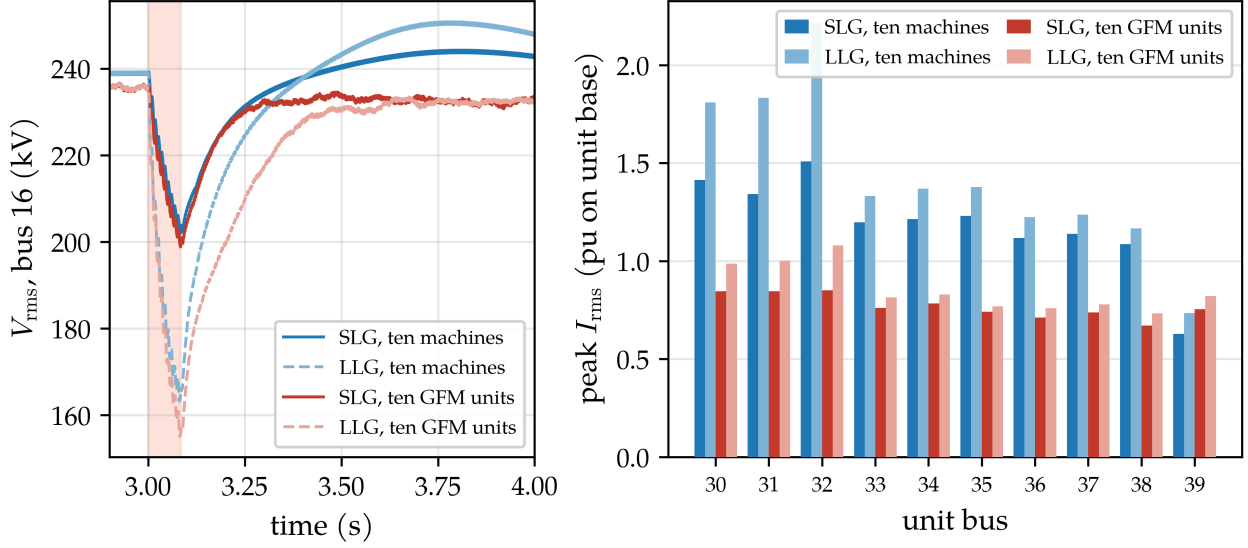


Figure 31. Unbalanced faults at bus 14, all-machine and all-GFM fleets, 10 s records with the reference timing and clearing. Left: bus-16 RMS voltage through the fault (SLG solid, LLG dashed). Right: peak RMS current of each unit on its own base, measured on the 230 kV side of its step-up transformer.

4 Deriving Dynamic Models of a Droop-Based GFM Inverter and a Synchronous Machine

The simulations in Section 3 show fault ride through and contrasting post-fault responses. To interpret those differences, we compare the dynamic equations of the machine and the droop-controlled inverter. Equivalent inertia opposes frequency acceleration, while equivalent damping opposes frequency deviation. These reduced descriptions connect device controls to the responses observed in the network.

A synchronous machine stores kinetic energy in its rotor, whereas an inverter generates its electrical phase through control. The two phase states can occupy analogous positions in reduced equations without representing the same physical mechanism. An inverter draws energy from its DC source and circuit storage; an equivalent inertia coefficient does not imply a rotating energy store. The analogy is useful within its assumptions, not a replacement for network, voltage-control, and limiter dynamics.

Prior work develops droop-converter stability analysis with current limitation [18], examines the applicability of equal-area reasoning [19], and describes equivalent-circuit models [20]. Interaction-energy methods extend the analysis to multiple GFM inverters [21]. We use the swing-form development in [1] to compare the selected models. The following derivation establishes consistent frequency normalization before discussing the effects of limiting.

4.1 Synchronous Machine Dynamic Model

The classical model relates electrical power imbalance to rotor acceleration [22, 23]. Let $\omega_0 = 2\pi f_n$ denote nominal electrical angular frequency in rad/s, where f_n is nominal frequency in Hz. Define the normalized frequency deviation as $\Delta\bar{\omega} = (\omega_e - \omega_0)/\omega_0$, in per unit (the bar marks a per-unit quantity). Let θ_E be the phase of the internal voltage and θ_V that of the grid voltage, both measured from the synchronous reference; the electrical phase displacement that sets the power transfer is the angle across the reactance, $\delta = \theta_E - \theta_V$. With the stiff grid as the reference, θ_V is constant and

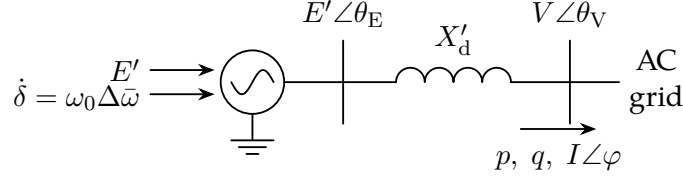


Figure 32. Classical synchronous-machine model: transient internal voltage $E' \angle \theta_E$ behind reactance X'_d , connected to the grid at $V \angle \theta_V$. The internal magnitude is set by the field circuit. The displacement $\delta = \theta_E - \theta_V$ is tied to rotor motion and satisfies $\dot{\delta} = \omega_0 \Delta \bar{\omega}$.

$\dot{\delta} = \dot{\theta}_E = \omega_0 \Delta \bar{\omega}$. With powers normalized on the machine apparent-power base,

$$\frac{2H}{\omega_0} \ddot{\delta} + \frac{D}{\omega_0} \dot{\delta} = p_m - p_e, \quad (1)$$

where H is the inertia constant in seconds and D is normalized power per normalized frequency deviation. Mechanical input and electrical output are p_m and p_e , respectively. The physical kinetic-energy definition is $H = J\omega_{m0}^2 / (2S_{\text{base}})$, where J is moment of inertia in kg m^2 , ω_{m0} is nominal mechanical speed in rad/s, and S_{base} is in VA. Electrical angular frequency equals mechanical angular speed multiplied by the number of pole pairs.

For constant voltage magnitudes connected through total reactance X_{tot} , the electrical power is $p_e = p_{\text{max}} \sin \delta$. Here $p_{\text{max}} = E'V/X_{\text{tot}}$, with transient internal voltage E' , grid voltage V , and reactance expressed on compatible per-unit bases. Taking the stiff grid as the angle reference gives

$$M \ddot{\delta} = p_m - p_{\text{max}} \sin \delta - \frac{D}{\omega_0} \dot{\delta}, \quad M = \frac{2H}{\omega_0}. \quad (2)$$

Figure 32 shows this single-machine infinite-bus representation.

4.2 GFM Inverter Dynamic Model

The simplified GFM model used in the analysis is a controllable voltage source behind coupling reactance X_L [10]. Its topology resembles the classical machine circuit, but the controller determines its internal voltage magnitude and phase. Figure 33 shows the interface and Figure 34 shows the corresponding Norton equivalent used in a network solution. The equivalent circuit does not change the resource's GFM control classification.

The controlled source has magnitude E and phase θ_E , while the terminal grid voltage has magnitude V and phase θ_V ; as for the machine, the displacement $\delta = \theta_E - \theta_V$ across the coupling reactance sets the power transfer. In the machine representation, the field circuit sets the internal voltage magnitude and rotor motion determines its phase. In the GFM representation, the power and voltage controls perform the analogous functions. The controller acts on filtered power measurements rather than directly applying its droop gain to instantaneous unfiltered power.

In the analyzed runs the REGFM_A1 controls are the P - f droop of Equation (3), whose frequency is integrated to the internal phase θ_E ; the Q - V droop with the voltage regulator ($k_{pv} = 0$, $k_{iv} = 5.86$ pu/s, $E_{\text{max}} = 1.15$, $V_{\text{Flag}} = 1$, regulating the point-of-interconnection voltage); the reactive limits $Q_{\text{max}} = -Q_{\text{min}} = 0.44$; the overload controller ($k_{pp\text{max}} = 0.01$, $k_{ip\text{max}} = 0.1$ pu/s), which acts only while filtered power exceeds P_{max} ($P_{\text{max}} = 0.9$, $P_{\text{min}} = 0$ in the single-device and inertia-test cases; $P_{\text{max}} = 2.0$, $P_{\text{min}} = -0.9$ in the 39-bus cases, where it never engages); and the transient current limit $I_{\text{max}} F$. The plant-controller references P_{ref} , Q_{ref} , and V_{ref} are held at their initialized values. The model exposes no inertia constant (Figure 9); its equivalent inertia

is the filtered-droop quantity $T_{Pf}/2m_p$ of Equation (5) only. No secondary control, that is, no slower loop that returns frequency and voltage to their reference values after the droop response (plant-level integral control or automatic generation control), acts in any run [10]. The post-fault operating points are the droop equilibria, and the effect of such a loop on stability is not assessed here. Figures 35 and 36 show the machine governor path and the inverter controls as configured.

Figure 35 reads as follows. The speed error $P_{ref} - \omega - R\tilde{P}_e$, with $R_{select} = 1$ selecting the electrical power P_e filtered by T_{pelec} , is clipped to $[\text{minerr}, \text{maxerr}] = \pm 0.055$ per unit, so a speed error larger than 0.055 per unit (3.3 Hz) produces no further governor action. The clipped error drives the proportional gain and the integral branch; the integrator input $(K_{igov}/K_{pgov})(fsr - s_2)$ equals K_{igov} times the error while the governor is selected and tracks fsr otherwise, which is the PES-TR1 anti-windup implementation of the PI controller. The load limiter (L_{dref} , K_{pload} , K_{iload} , with the exhaust-temperature path through T_{sa} , T_{sb} , and T_{load}) and the acceleration limiter (A_{set} , K_a , T_a) use the same tracking structure. The Low Value Select block compares the three fuel demands, governor ($fsrn$), load limiter ($fsrt$), and acceleration limiter ($fsra$), and passes the smallest to the valve as fsr , so whichever controller asks for the least fuel is in control. The valve stroke is rate-limited (R_{open} , R_{close}) and position-limited (v_{min} , v_{max}); with $Flag = 1$ the fuel flow Wf is the stroke times speed, and $P_{mech} = K_{turb}(Wf - W_{fnl})$ through the T_c , T_b lead-lag. The only integral term in the governor path is K_{igov} .

Figure 36 redraws the outer controls from Figures 3(a), 3(b), and 4 of the specification [10]; its inputs P_{inv} , Q_{inv} , and V_{inv} have already passed through the T_{Pf} , T_{Qf} , and T_{Vf} measurement filters. In the P - f droop, $\Delta\omega = \omega_0 [m_p(P_{ref} - P_{inv}) + \Delta\omega_{OL}]$, which is $\omega_0\Delta\bar{\omega}$ of Equation (3) with the overload term added, is integrated to δ_{droop} ; the overload mitigation term $\Delta\omega_{OL}$ is nonzero only above P_{max} or below P_{min} because its two branches are limited to non-positive and non-negative values. In the Q - V droop, $V_{flag} = 1$ selects the proportional-integral voltage regulator that drives the point-of-interconnection voltage V_{inv} to V_{cmd} , with its output limited to $[E_{min}, E_{max}] = [0, 1.15]$; the Q_{max}/Q_{min} branches add to V_{cmd} only beyond the ± 0.44 per-unit limits. In the fault current limiter, while $I < I_{max F}$ the internal voltage is the droop output; once $I \geq I_{max F}$ it is recomputed from the terminal voltage and the coupling reactance so that the current magnitude is held at $I_{max F}$ with its phase φ unchanged. The integral terms are k_{iv} , k_{ipmax} , and k_{iqmax} .

Active-power droop sets frequency, while reactive-power droop adjusts voltage magnitude. For the unsaturated reduced model, the normalized laws are

$$\Delta\bar{\omega} = m_p(p^* - \tilde{p}), \quad E = E_0 + m_q(q^* - \tilde{q}), \quad (3)$$

where p^* , q^* are power references and $\tilde{p}$, $\tilde{q}$ are filtered power measurements. The active-power filter satisfies

$$T_{Pf}\dot{\tilde{p}} = p - \tilde{p}, \quad (4)$$

where p is the unfiltered normalized active power. The controller-generated phase satisfies $\dot{\theta}_E = \omega_0\Delta\bar{\omega}$ relative to the same synchronous reference, so against the stiff grid $\dot{\delta} = \omega_0\Delta\bar{\omega}$ as for the machine. Table 3 defines the droop variables and their units.

4.3 Modeling GFM Equivalent Inertia and Equivalent Damping Dynamics

For a fixed active-power reference, differentiating the first droop law relates frequency acceleration to the filtered-power derivative. Substituting Equation (4) eliminates that derivative. Rearranging

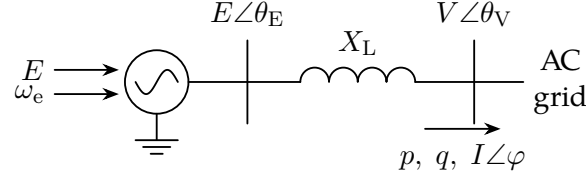


Figure 33. Simplified inverter model used in the analysis, after Figure 1 of the REGFM_A1 specification [10]. A controllable voltage source $E\angle\theta_E$ behind coupling reactance X_L delivers active and reactive power to the grid at $V\angle\theta_V$. Under unsaturated filtered droop, $\dot{\theta}_E = \omega_0 \Delta\bar{\omega}$.

the power terms gives

$$\begin{aligned} \frac{d\Delta\bar{\omega}}{dt} &= -m_p \dot{p} = -\frac{m_p}{T_{Pf}} (p - \tilde{p}) \\ &= -\frac{m_p}{T_{Pf}} (p - p^*) - \frac{\Delta\bar{\omega}}{T_{Pf}}. \end{aligned}$$

Multiplication by T_{Pf}/m_p then gives

$$\underbrace{\frac{T_{Pf}}{m_p} \frac{d\Delta\bar{\omega}}{dt}}_{2H_{eq}} = p^* - p - \underbrace{\frac{1}{m_p} \Delta\bar{\omega}}_{D_{eq}}, \quad \boxed{H_{eq} = \frac{T_{Pf}}{2m_p}, \quad D_{eq} = \frac{1}{m_p}}. \quad (5)$$

In terms of the displacement $\delta = \theta_E - \theta_V$, this becomes

$$\frac{2H_{eq}}{\omega_0} \ddot{\delta} + \frac{D_{eq}}{\omega_0} \dot{\delta} = p^* - p. \quad (6)$$

Both coefficients therefore have the same normalization as in Equation (1).

The active-power filter creates equivalent inertia: a longer filter time constant increases H_{eq} , while H_{eq} vanishes as that time constant approaches zero. A smaller droop gain increases both equivalent inertia and equivalent damping when the filter time constant is fixed. These statements apply to the unsaturated filtered-droop path with a fixed reference, not all inner-loop or hardware dynamics. Energy-based inertia tests, including those discussed in IEEE 2800 Annex K.2.2 and UNIFI Section 5.6 [25, 5], require care because damping can also contribute to the measured power response.

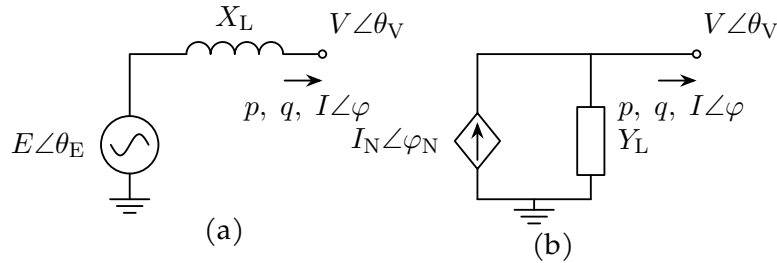


Figure 34. Equivalent normalized circuits, after external Figure 2 of [10]: internal voltage source behind reactance and its Norton representation. Here $I_N\angle\varphi_N$ is the equivalent injected current and $Y_L = 1/(jX_L)$ the parallel admittance; φ denotes current phase. Solver representation does not determine GFM or GFL classification.

Table 3. Variables in Equations (3) and (4).

Symbol	Meaning	Units
ω_0	Nominal electrical angular frequency	rad/s
$\Delta\bar{\omega}$	Normalized (per-unit) electrical frequency deviation	p.u.
E_0	Nominal internal-voltage reference	p.u.
p^*, q^*	Active/reactive power references	p.u.
$\tilde{p}, \tilde{q}$	Low-pass-filtered power measurements	p.u.
$\dot{\tilde{p}}$	Filtered active-power time derivative	p.u./s
m_p, m_q	Frequency/power and voltage/reactive-power droop gains	p.u./p.u.
T_{Pf}	Active-power measurement-filter time constant	s

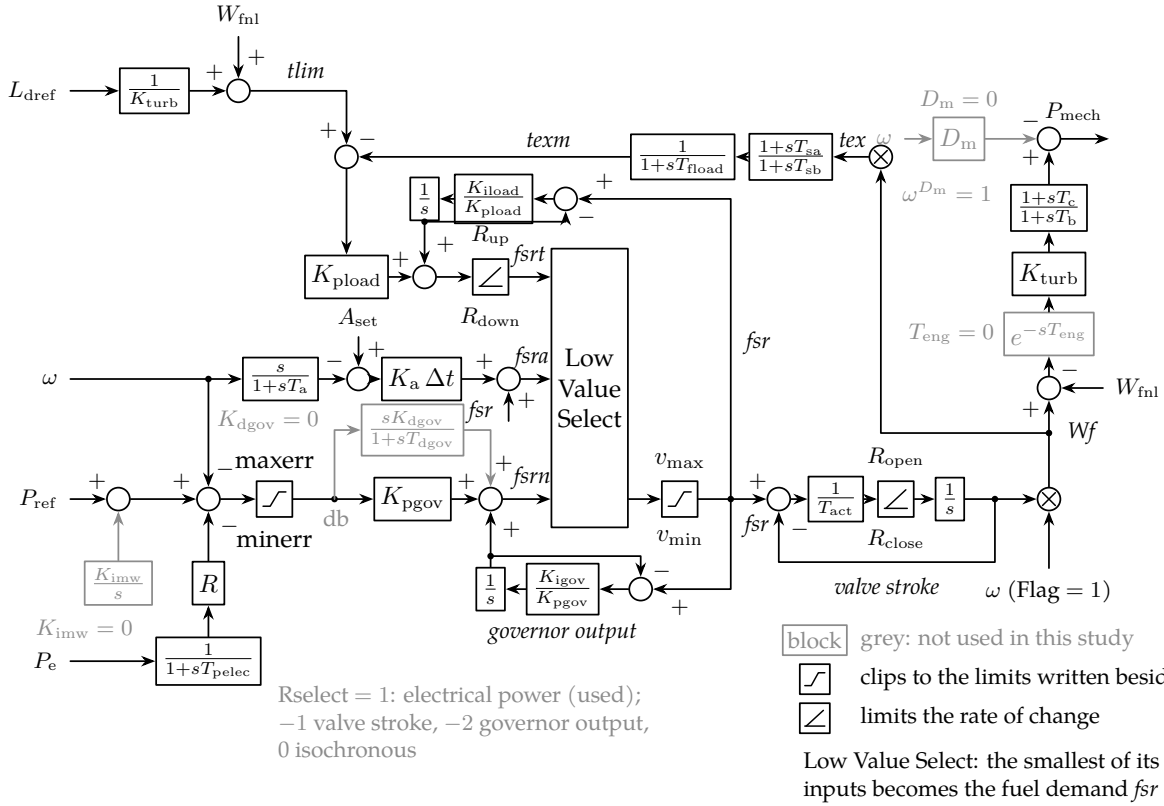


Figure 35. GGOV1 governor and turbine model of the machine, redrawn block for block from Figure 3-5 of IEEE PES-TR1 [40] with the Table 1 settings and the figure's signal names; the signal path is described in Section 4.2. Grey blocks and wires are present in GGOV1 but switched off by the settings ($K_{imw} = K_{dgv} = T_{eng} = D_m = 0$) and carry no signal in any run.

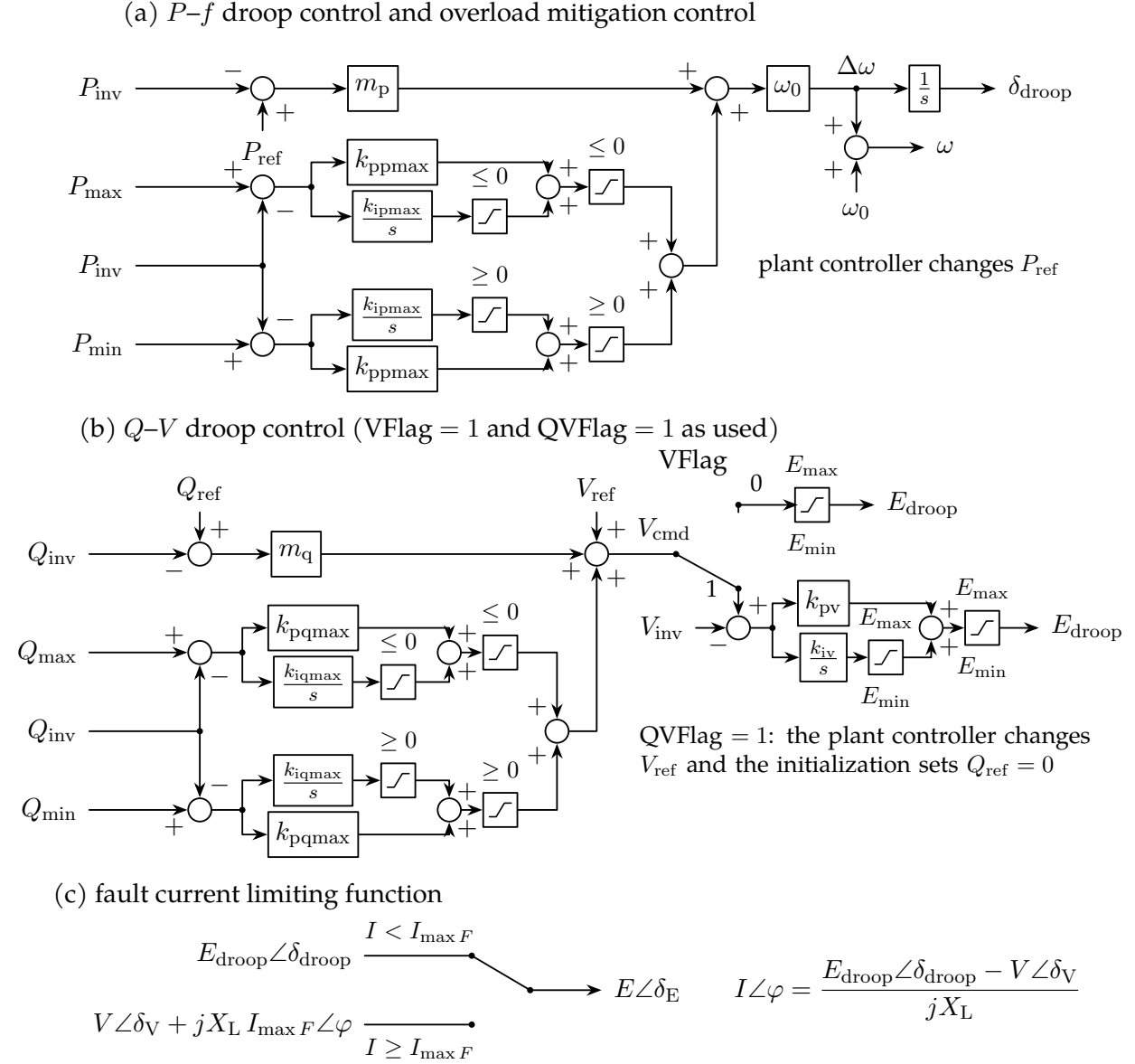


Figure 36. REGFM_A1 outer controls, redrawn from Figures 3(a), 3(b), and 4 of the model specification [10]: (a) P - f droop with overload mitigation, (b) Q - V droop with the voltage regulator selected by VFlag = 1, and (c) fault current limiting. The inputs have passed through the T_{Pf} , T_{Qf} , and T_{Vf} measurement filters, and the droop output $\Delta\omega = \omega_0 \Delta\bar{\omega}$ is in rad/s.

4.4 Current Limiting Dynamic Model

Semiconductor current and thermal limits motivate fast current protection [38]. The REGFM_A1 model constrains its current demand through the limiter described in [10]; the reference setting is 2.0 per unit. A simple circuit illustrates why increasing angular separation can activate that

constraint. For two equal voltage magnitudes E separated by angle δ across positive reactance X_{tot} ,

$$\begin{aligned} I &= \frac{E}{jX_{\text{tot}}}(e^{j\delta} - 1), \\ e^{j\delta} - 1 &= 2je^{j\delta/2}\sin(\delta/2), \\ |I| &= \frac{2E}{X_{\text{tot}}}|\sin(\delta/2)|. \end{aligned} \quad (7)$$

Here j is the imaginary unit, and I is the current phasor on the compatible circuit base. Current demand increases with angle on $0 \leq \delta \leq \pi$ until it encounters the limit I_{max} . Equal voltages and a purely reactive connection make this an illustration, not a general deep-fault model.

4.5 Single Machine Infinite Bus Modeled Fault Response

Figures 37 and 38 compare the selected machine and GFM models under the same single-device bolted fault. The machine delivers approximately 7–8 per-unit fault current, influenced by its subtransient reactance and the external network. Its post-fault active power oscillates at 1.5 Hz with a decay rate of 1.6 s^{-1} (damping ratio 0.17, from the log decrement of successive peaks after 4.4 s), so the oscillation falls below 1% of its first swing about 3 s after clearing, whereas the inverter's active power returns to its reference within approximately 0.5 s: the infinite bus holds frequency, so Equation (3) returns $\tilde{p}$ to p^* as $\Delta\bar{\omega} \rightarrow 0$. In the 39-bus fleets no restoration loop exists, and frequency and power settle at a droop-shared post-fault operating point. The GFM current setting is 2.0 per unit in this comparison, distinct from the 1.5 per-unit bus-10 study.

The fault is applied at the point of interconnection on the grid side of each device. The machine, 2000 MVA, is modeled at 230 kV without a step-up transformer; the inverter case uses the release configuration, 0.1 MVA at 0.48 kV, with the source at the same voltage. Each grid is an infinite bus behind SCR 10 and $X/R = 20$ on the device base. The inverter dispatch is 0.50 per unit, matching the machine's measured 0.50 per unit; pre-fault reactive power is 0.30 per unit for the machine at 1.03 per-unit terminal voltage, set by its exciter, and -0.02 per unit for the inverter at 1.00 per unit.

Figure 39 separates the active and reactive current contributions. Both active powers return to 0.50 per unit; the active-current levels differ after clearing because $I_P = P/V$ and the machine's terminal voltage settles at 1.06 per unit under its exciter while the inverter's is 1.00 per unit. The selected GFM model provides a rapid reactive-current response, 0.4–0.6 per unit through the first 50 ms after clearing. The machine response includes field-circuit dynamics: its reactive current is negative for 0.16 s after clearing, reaching -0.65 per unit at 3.12 s while its terminal voltage recovers from 0.95 to 1.02 per unit, and its exciter then raises the reactive power to 0.59 per unit at 1.06 per-unit voltage by 5 s. This comparison connects voltage support to device structure and controls. It does not imply identical current capabilities or response bandwidths across inverter designs.

4.6 Equivalent Damping, Inertia, and Settling Time

Both models share the swing form, so their damping coefficients can be compared directly. For small deviations of the per-unit speed, Equation (2) reads

$$2H \frac{d\Delta\bar{\omega}}{dt} = \Delta P_{\text{mech}} - \Delta P_e - D \Delta\bar{\omega}, \quad (8)$$

where ΔP_{mech} and ΔP_e are the deviations of the mechanical and electrical power from their pre-disturbance values, and the damping term is the power that opposes the speed deviation. A device whose control changes its power in response to speed therefore contributes $D = -\Delta P / \Delta\bar{\omega}$, where ΔP is that change: the governor's change in mechanical power, ΔP_{mech} , for the machine, and the

droop's change in output power for the inverter. D is positive when the power falls as the speed rises. For a sinusoidal speed swing only the part of the power response in step with the speed acts as damping, the damping-torque component of [24]; the part a quarter cycle out of step is in step with the acceleration and acts as inertia.

For the inverter, Equation (5) gives $D_{eq} = 1/m_p = 100$ for $m_p = 0.01$, the same at every frequency within the filtered-droop model. For the machine, the governor-and-turbine path gives a frequency-dependent coefficient, the term $D + \text{Re}\{G(j\Omega)\}/R_{SG}$ of the synchronous-generator operator in [1, Eq. (50)], where G is the normalized response from speed deviation to mechanical power ($G(0) = 1$), R_{SG} the permanent droop, and $\Omega = 2\pi f$. Evaluated from the GGOV1 blocks of Figure 35 with the Table 1 settings,

$$\frac{G(s)}{R_{SG}} = -\frac{\Delta P_{mech}}{\Delta \bar{\omega}} = \frac{K_{turb} T(s) [A(s) C(s) (1 - 2H R_{SG} s F(s)) - W_{f0}]}{1 + R_{SG} F(s) K_{turb} T(s) A(s) C(s)}, \quad (9)$$

where $C(s) = K_{pgov} + K_{igov}/s$ is the governor PI controller ($K_{dgv} = 0$); $A(s) = 1/(1 + sT_{act})$ is the actuator, a first-order lag; $T(s) = (1 + sT_c)/(1 + sT_b)$ is the turbine lead-lag; $F(s) = 1/(1 + sT_{pelec})$ is the first-order lag of the electrical-power measurement selected by $R_{select} = 1$; and $W_{f0} = W_{fnl} + P_{m0}/K_{turb}$ is the pre-disturbance fuel flow at dispatch P_{m0} , which enters because $\text{Flag} = 1$ makes the fuel flow the product of valve stroke and speed. The power fed back through $F(s)$ is that of a machine swinging against a stiff network, $\Delta P_e = \Delta P_{mech} - 2Hs \Delta \bar{\omega}$, which gives the factor $(1 - 2H R_{SG} s F)$ with $H = 5.46$ s; the factor is 1 at zero frequency and $0.47 - 0.09j$ at 1 Hz, where it removes about half of the direct speed path. In the 39-bus fleet ΔP_e also carries the synchronizing

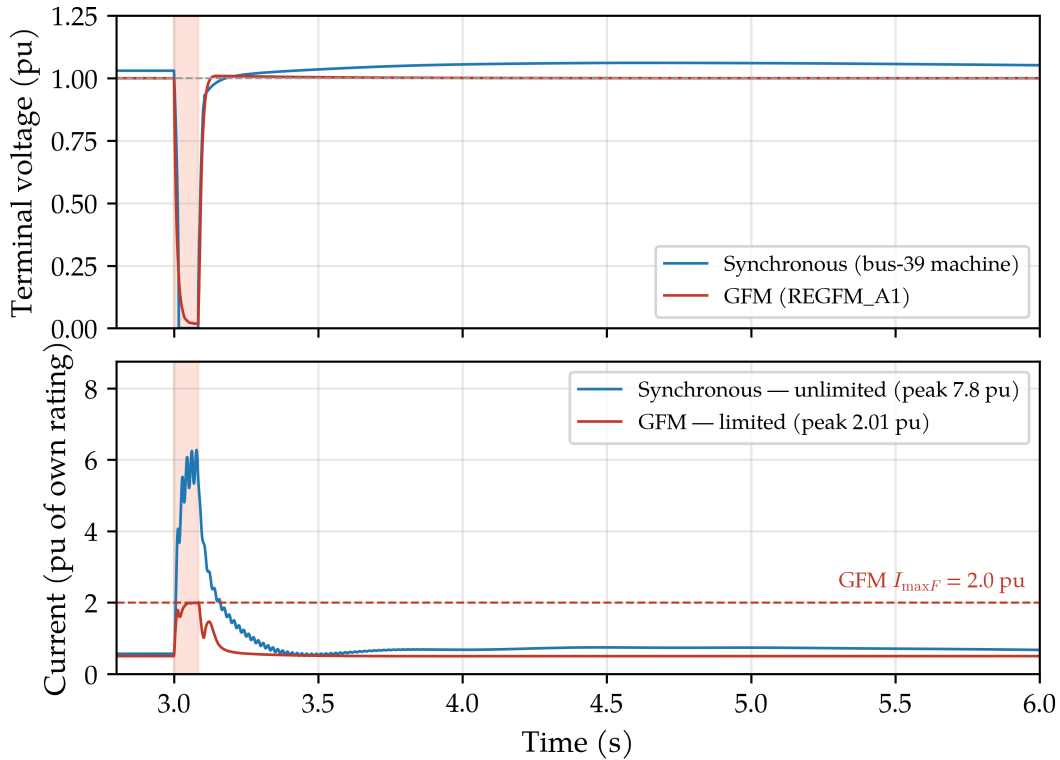


Figure 37. Single-device fault response: voltage and current at the device terminals for the synchronous machine and GFM inverter; neither single-device case has a step-up transformer. The GFM transient-current setting is 2.0 per unit; both devices are dispatched at 0.50 per unit.

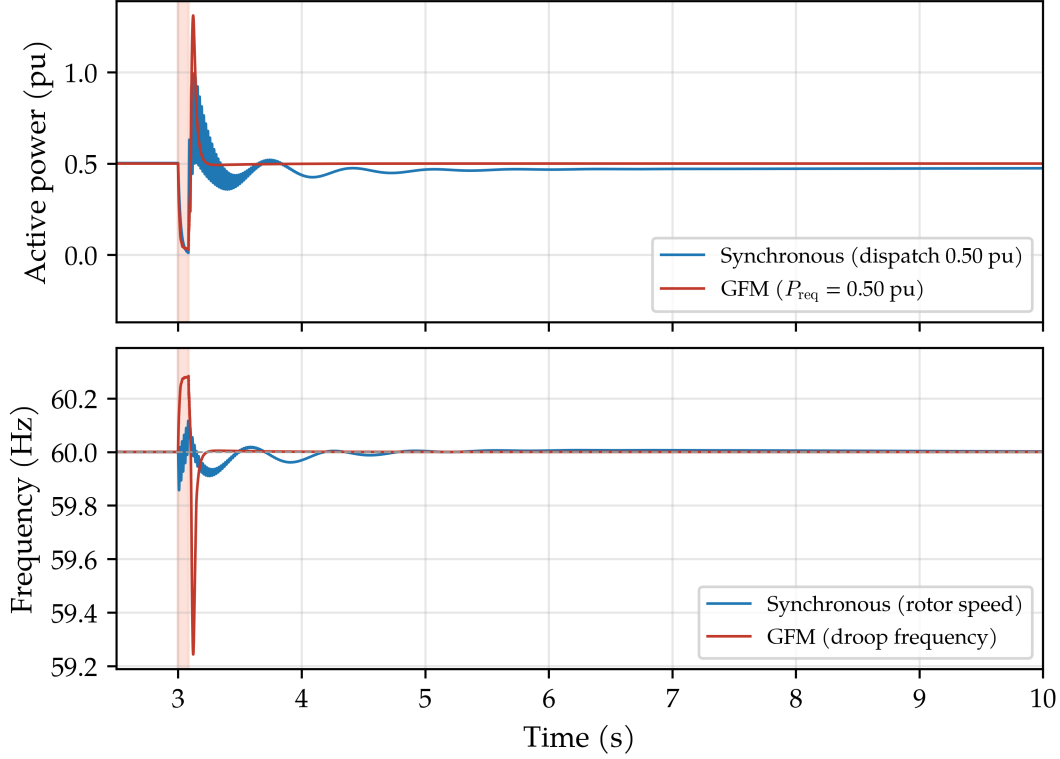


Figure 38. Single-device active-power and frequency responses at matched 0.50 per-unit dispatch. The selected GFM model settles faster than the machine case. The dense band on the machine trace for about 0.5 s after clearing is a 60 Hz ripple of the instantaneous power calculation, produced by the direct-current offset of the armature current after the bolted fault; it decays with that offset and is not an oscillation of the machine. The machine's 1.5 Hz oscillation that follows decays at 1.6 s^{-1} (Section 4.5).

power exchanged with the other machines, so Equation (9) is the single-machine form. $D = 0$ in the GENROU record.

At zero frequency the path gives $1/R_{SG} = 20$, so the five-to-one comparison $R_{SG}/m_p = 0.05/0.01$ holds for these normalized coefficients in the steady-state limit only.

Figure 40 is an analytical estimate. It plots Equation (9), evaluated from the governor and turbine constants of Table 1 alone, over the 0.1 to 3 Hz band of electromechanical swings for the fleet's dispatch range, $P_{m0} = 0.50$ to 1.00 per unit of each machine's rating (bus 39 at 0.50, the other nine at 0.90–1.00). The governor holds speed by cutting fuel when the machine speeds up and adding fuel when it slows, so for a slow change its power change opposes the speed deviation and damps the swing; the actuator and the turbine delay that response. The coefficient starts at 20, falls as the two lags accumulate phase, crosses zero between 0.38 and 0.41 Hz, and reaches -1.4 to -1.8 at 1 Hz, the frequency of the inter-machine swing in Figure 23. Past the crossing the delay exceeds a quarter of the swing period, so the commanded power change arrives after the speed has reversed: mechanical power rises while the speed rises and feeds the swing. This is negative damping in the sense of [23, Sec. 2.1, Eq. (2.2); Sec. 12.5]. The sign was confirmed in PSCAD on the single-machine case of Section 4.5 by modulating the infinite-bus frequency: the measured coefficient is negative from 0.5 to 2 Hz at both dispatches. The sign change needs both lags, $T_{act} = 0.5 \text{ s}$ and $T_b = 0.1 \text{ s}$; a single first-order lag, the stand-in of [1, Eq. (51)], never reaches 90° and cannot change sign. Closing the feedback with $\Delta P_e = \Delta P_{mech}$ instead, or omitting the speed dependence of the fuel flow, moves the crossing to 0.6–0.7 Hz and the 1 Hz value to between

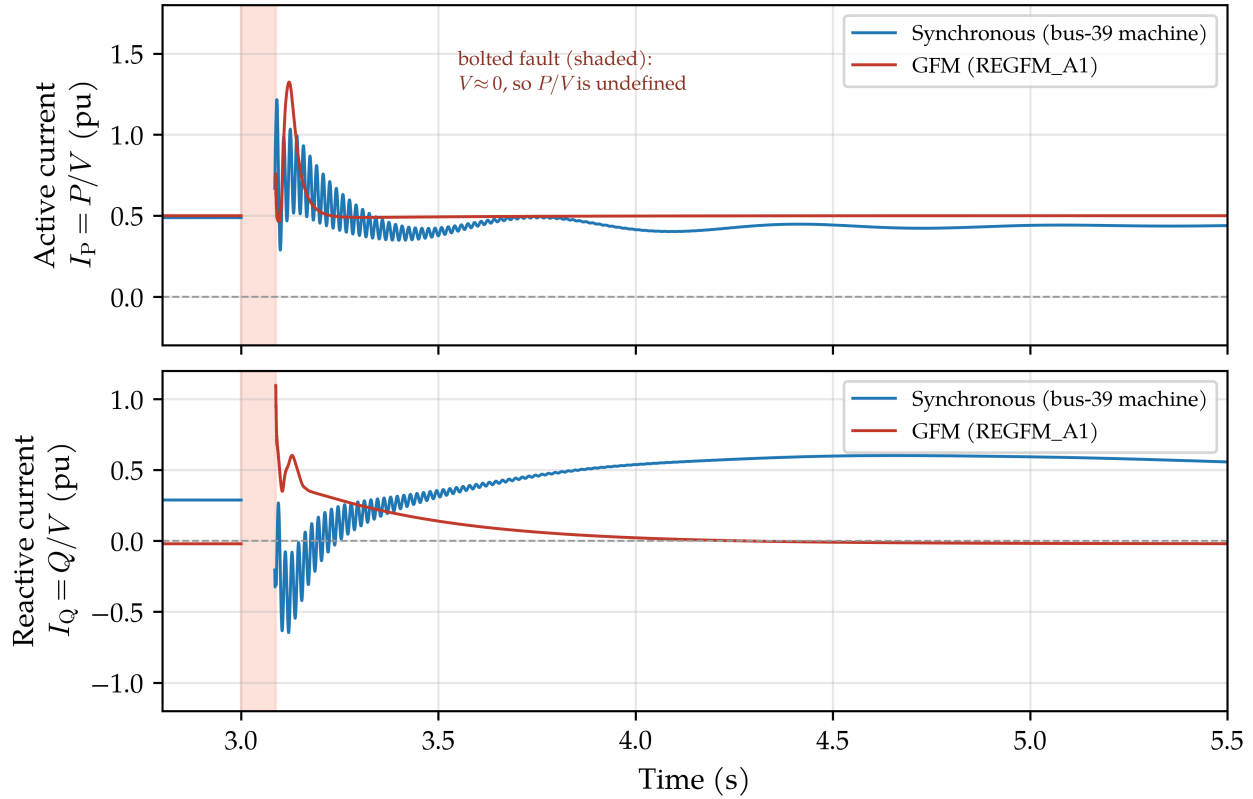


Figure 39. Active and reactive current components for the single-device fault comparison, $I_P = P/V$ and $I_Q = Q/V$ on each device base, positive for injection into the grid and blanked while $V < 0.2$ per unit; both devices are dispatched at 0.50 per unit. The 60 Hz ripple on the machine traces is the armature direct-current offset described for Figure 38.

−1.1 and −2.0 without changing the sign. Damper windings, excitation, the stabilizer, and network interactions also damp a mode, and none of them is in Equation (9); PSS2B and ESST4B are active in all machine cases, and no case without the stabilizer was simulated. The inverter’s coefficient, by contrast, is flat where the machine’s changes sign. No simulation result enters the figure or Table 4; both follow from the device parameters.

The inertia comparison runs the other way: $T_{Pf} = 10$ ms and $m_p = 0.01$ give $H_{eq} = 0.5$ s, roughly one eleventh of the machine’s 5.46 s. Table 4 collects the values. Strong equivalent damping with low equivalent inertia is consistent with the faster GFM settling.

Table 4. Selected inertia and damping quantities, all analytical values from the device parameters (Equations (5) and (9)). Damping coefficients use normalized power per normalized frequency.

Quantity	GFM (droop)	Synchronous machine
Inertia constant	$H_{eq} = 0.5$ s	$H = 5.46$ s
Damping coefficient, zero frequency	100	20 (governor)
In-phase coefficient, 1 Hz	100	−1.4 to −1.8 (governor)
Governor zero crossing	Not applicable	0.38–0.41 Hz

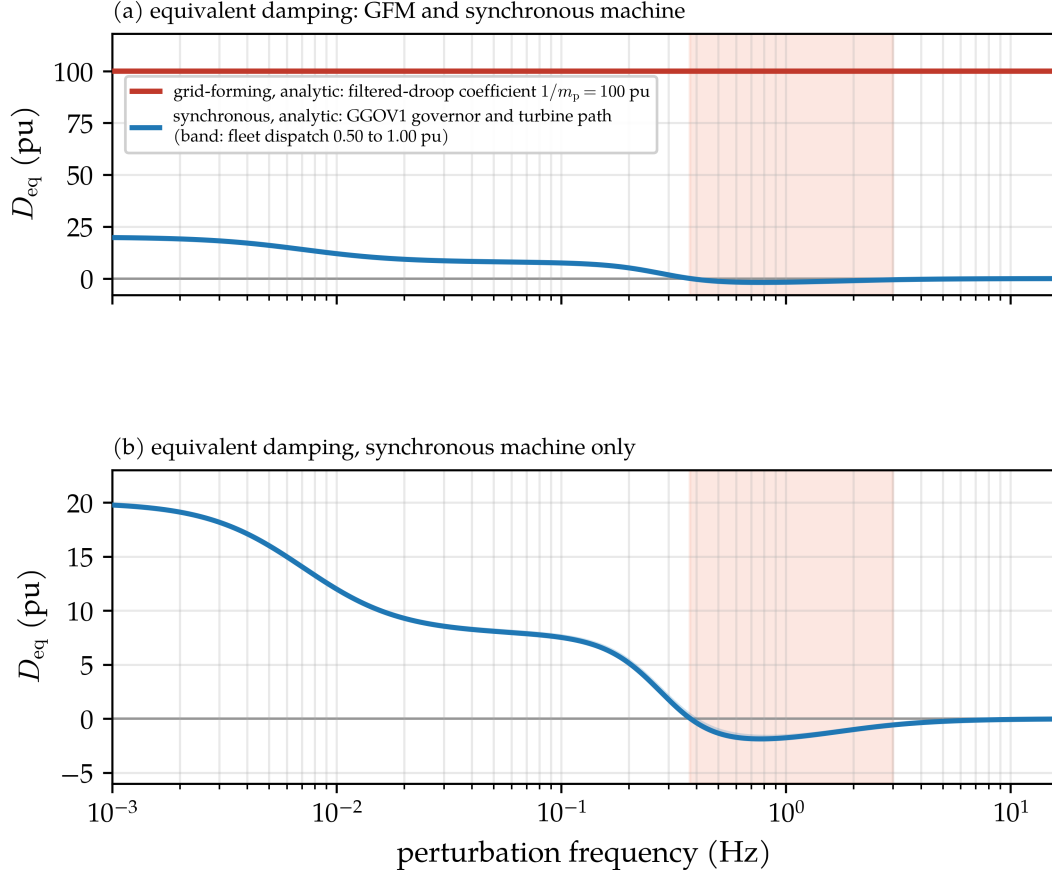


Figure 40. Analytical estimate of the equivalent damping versus perturbation frequency, normalized power per normalized frequency: the GFM filtered-droop coefficient $D_{eq} = 1/m_p = 100$ and the machine governor-and-turbine path of Equation (9) (band: fleet dispatch 0.50 to 1.00 per unit; line: median dispatch; shading: where the path is negative).

4.7 Measuring GFM Equivalent Inertia

The ERCOT PMView inertia estimation test applies a frequency ramp and calculates an estimated equivalent inertia H_{est} from the incremental power response. We used PMView 3.5 profile 17, associated with the Dynamics Working Group Procedure Manual Revision 24, Section 3.1.5.12 [14, 15]. The test uses short-circuit ratio (SCR) 3, reactance-to-resistance ratio $X/R = 6$, and a -1 Hz/s source-frequency ramp at nominal frequency 60 Hz. Over the 0.5 s integration window,

$$H_{est} = 60\Delta E, \quad \Delta E = \int_0^T (p - p_0) dt, \quad T = 0.5 \text{ s}, \quad (10)$$

where p_0 is pre-disturbance power and positive incremental power denotes additional injection. Normalized power must use the same base as the inertia comparison; ΔE and H_{est} then have units of seconds. The imposed source frequency and the inverter's internal frequency need not coincide during the network transient.

For the synchronous-machine case, the test gives $H_{est} = 5.20$ s against the configured inertia of 5.46 s, a difference of -4.8% . For the GFM case, it gives 7.44 s against the filtered-droop equivalent inertia of 0.5 s, approximately fourteen times larger. Figures 45 and 46 show these responses. Increasing SCR from 3 to 20 raises the GFM estimate to 10.06 s without changing the configured droop parameters.

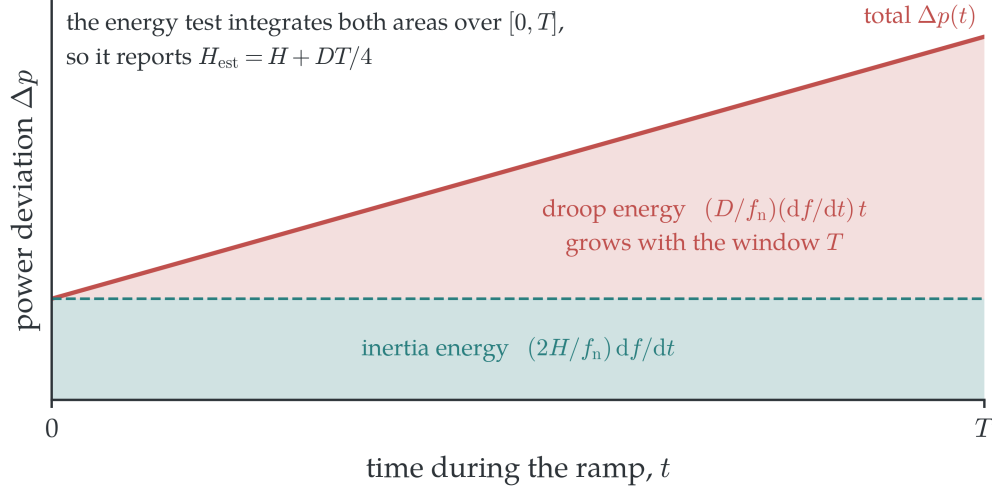


Figure 41. Contributions to the integrated power response during a constant rate-of-change-of-frequency ramp, from Equations (11) and (12). The inertial contribution is constant and forms a rectangle; the damping contribution accumulates and forms a triangle. The curves illustrate the analytical decomposition.

The damping contribution explains why the integrated estimate should not automatically be identified with equivalent inertia. Under the unsaturated swing-form model, assume that internal frequency follows the imposed constant ramp over the integration window. With $\Delta f(t) = \dot{f} t$, $\dot{f} = -1 \text{ Hz/s}$, and $f_n = 60 \text{ Hz}$, the incremental response is

$$\Delta p(t) = -\frac{2H}{f_n} \dot{f} - \frac{D}{f_n} \dot{f} t. \quad (11)$$

Integration gives

$$\Delta E = -\frac{2H}{f_n} \dot{f} T - \frac{D}{2f_n} \dot{f} T^2. \quad (12)$$

For a different window under the same ramp, the estimator multiplier scales from 60 to $30/T$, with T in seconds. Consequently,

$$H_{\text{est}} = H + \frac{DT}{4} = H + \frac{T}{4m_p}. \quad (13)$$

The last equality uses the filtered-droop damping coefficient, with $H = H_{\text{eq}}$.

The inertia term depends on how quickly frequency changes, whereas the damping term depends on how far frequency has moved from nominal. A constant ramp therefore produces different time dependences even though both contributions appear in the same active-power trace. Integrating that trace without separating the contributions assigns both to the inertia estimate. The distinction is between interpreting the test result and changing the configured device inertia, not between a resource with energy and one without energy.

The additional term shows that damping-related power contributes to the inertia estimate over a finite integration window. Figure 41 illustrates the constant inertial contribution as a rectangle and the accumulating damping contribution as a triangle. A longer window therefore increases the estimate even when device parameters remain unchanged. This reduced expression contains no network-strength dependence and does not, by itself, explain the SCR variation.

The machine and inverter also differ in response timescales. The inverter's 10 ms power filter allows droop response to develop within the half-second window, whereas turbine and governor

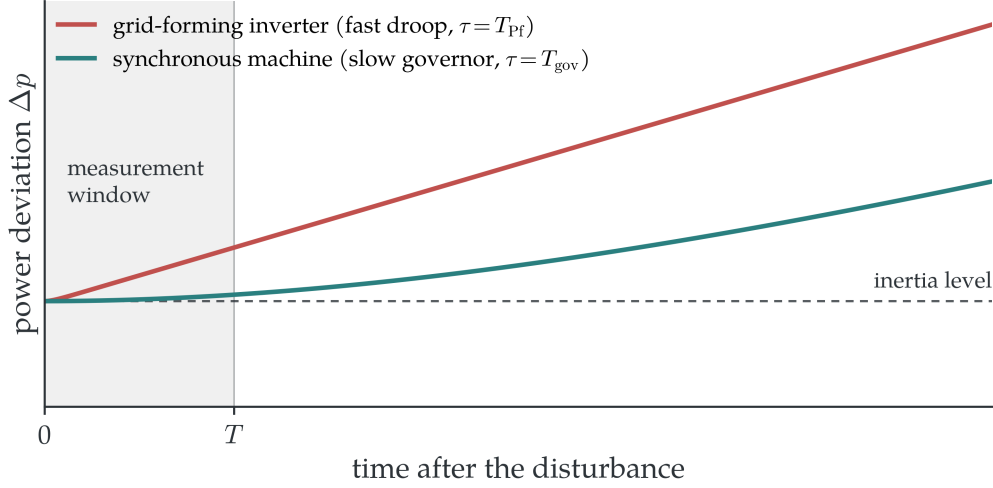


Figure 42. Illustrative time-scale separation between the inertial response and the droop response. The curves isolate the effect of droop-path lag: a faster path contributes within the measurement window, while a slower path contributes less over that interval.

dynamics delay the machine's contribution. Figure 42 illustrates that distinction. The machine estimate is close to configured inertia in this test, but its finite-window response need not be purely inertial.

Weakening the droop to $m_p = 0.1$ reduces the analytical equivalent inertia to 0.05 s and gives an estimated inertia of 1.22 s. Both equivalent inertia and damping change with this gain, so the difference is not exclusively a damping effect. Figure 48 shows the response, and Figure 43 compares the window sweep with Equation (13). The analytical estimates differ by approximately 14% at $T = 0.2$ s and 6% at $T = 0.5$ s; the line predicts the estimator output including damping, not an estimate with damping removed.

At $m_p = 0.01$, the unsaturated expression predicts 13.0 s rather than 7.44 s. The inertia-test run was dispatched at $P = 0$ with $P_{\max} = 0.9$ per unit (Figure 9), so 0.9 per unit of active-power headroom was available; the reduced model requests 0.85 per unit by 0.5 s, within that headroom, and the active-power limit therefore does not explain the difference. The measured power reached 0.69 per unit at the end of the window. The remainder is the network between the imposed source frequency and the inverter's internal frequency. Figure 44 isolates it: the test was repeated at short-circuit ratios from 2 to 40 with the inverter's internal droop frequency recorded. Referred to the imposed ramp, the estimate rises from 6.3 s at a short-circuit ratio of 2 to 10.4 s at 40, because the internal frequency follows the imposed -1 Hz/s ramp only partially through the coupling reactance, at 0.67 to 0.98 Hz/s. Referred to the internal frequency, the frequency the controller actually responds to, the same records give 9.4 to 10.6 s, and 9.8 to 10.4 s over the 3 to 20 range previously reported. The grid-strength dependence of the published estimate is therefore the network's lag between the two frequencies, not a change in the controller; the residual between 10 s and the analytic 13.0 s is the controller's own dynamics inside the window.

Equation (13) also suggests a damping correction. Since an underfrequency deviation is negative, removing the positive damping contribution requires

$$\Delta E_{\text{in}} = \int_0^T \left[\Delta p(t) + D \frac{\Delta f(t)}{f_n} \right] dt. \quad (14)$$

Applying the same test normalization to this corrected energy yields H . The correction requires

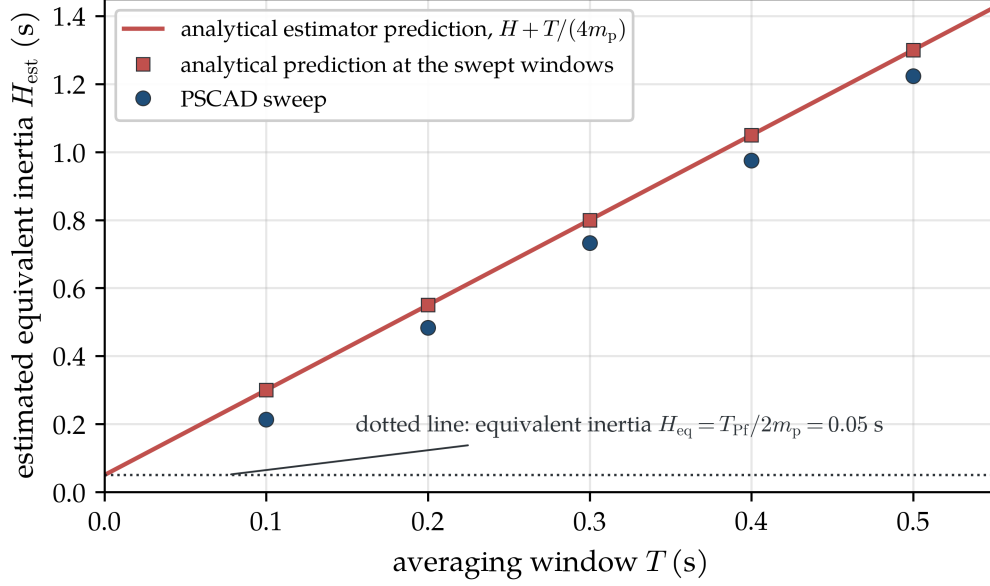


Figure 43. Inertia estimate from Equation (13) compared with the PSCAD integration-window sweep at $m_p = 0.1$. The intercept is the equivalent inertia $T_{Pf}/(2m_p) = 0.05$ s and the slope is $1/(4m_p) = 2.5$. The line predicts the inertia estimator output including damping, rather than an inertia estimate with damping removed. The dotted line is $H_{eq} = 0.05$ s.

Δp and D on the same per-unit base and Δf taken as the inverter's internal frequency, the one the droop responds to (Figure 44). It recovers H only where the reduced model holds: an active power limiter, the controller's own lag within the window, or the network's lag between the internal and imposed frequencies each leave a part of Δp that is neither inertial nor $D \Delta f$.

Alternatively, the window dependence is

$$H_{\text{est}}(T) = \underbrace{H}_{\text{intercept}} + \underbrace{\frac{1}{4m_p}}_{\text{slope}} T. \quad (15)$$

A window sweep could separate the intercept from the damping-related slope within this model. Figure 47 illustrates the dependence on integration window and control configuration. A disabled virtual-inertia parameter does not necessarily eliminate the equivalent inertia created by filtered droop.

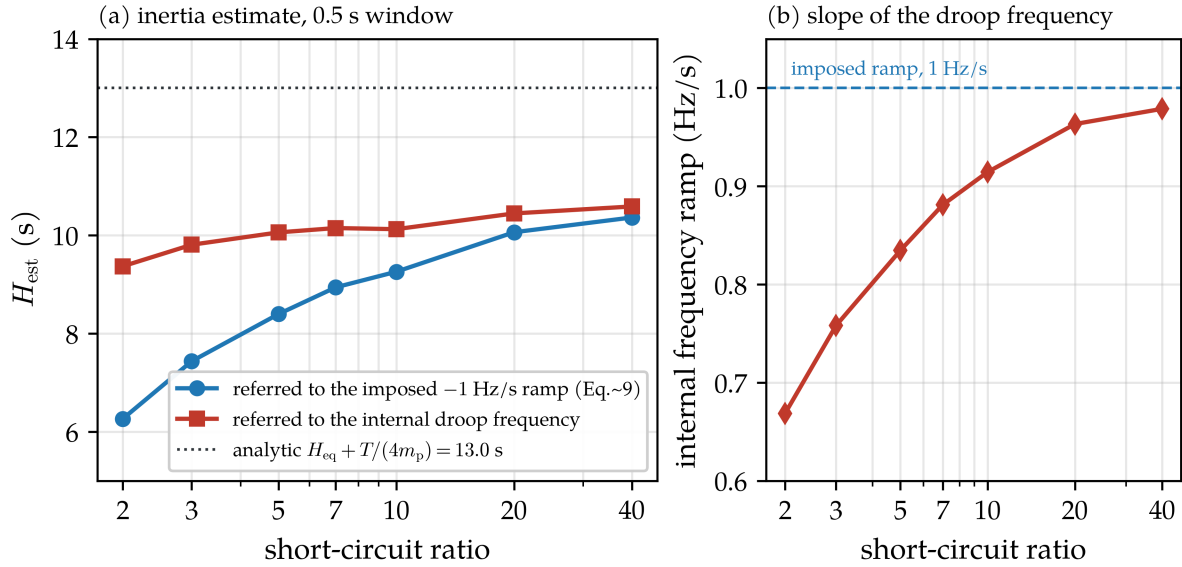


Figure 44. PMView inertia estimate of the droop-controlled inverter versus short-circuit ratio, 0.5 s window, dispatch $P = 0$. (a) The estimate referred to the imposed -1 Hz/s ramp, as in Equation (10), and referred to the inverter's internal droop frequency over the same window; the dotted line is the analytic $H_{\text{eq}} + T/(4m_p)$. (b) Least-squares slope of the internal droop frequency over the window. Short-circuit ratios 3, 5, 10, and 20 are the records behind the earlier 7.44 to 10.06 s range; 2, 7, and 40 were added for this figure.

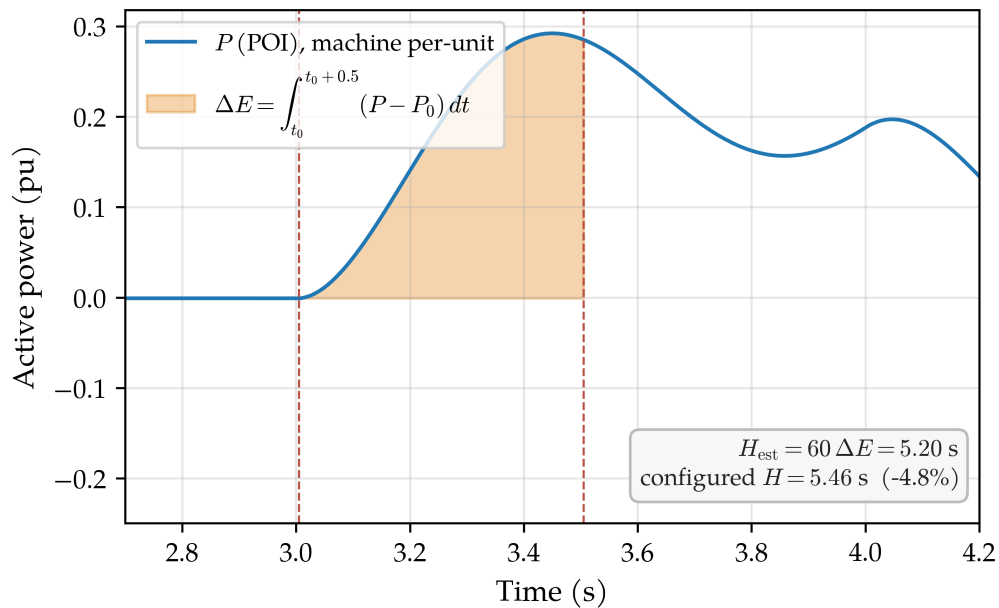


Figure 45. PMView inertia estimation test for the synchronous machine: $H_{\text{est}} = 5.20$ s against the configured inertia $H = 5.46$ s.

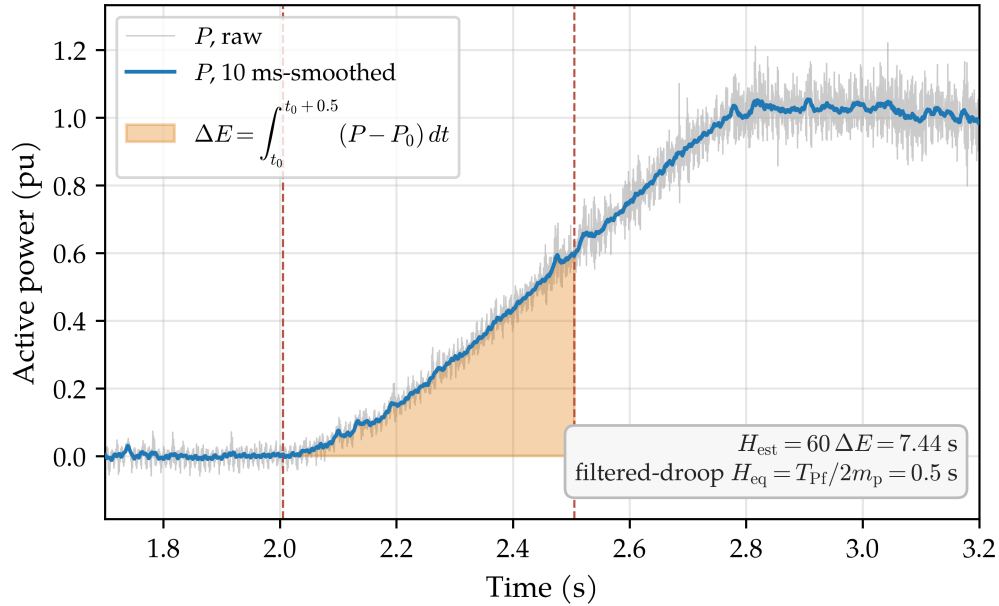


Figure 46. PMView inertia estimation test for the droop-controlled GFM inverter: $H_{\text{est}} = 7.44 \text{ s}$. This estimate includes the controller response and is distinct from the filtered-droop equivalent inertia.

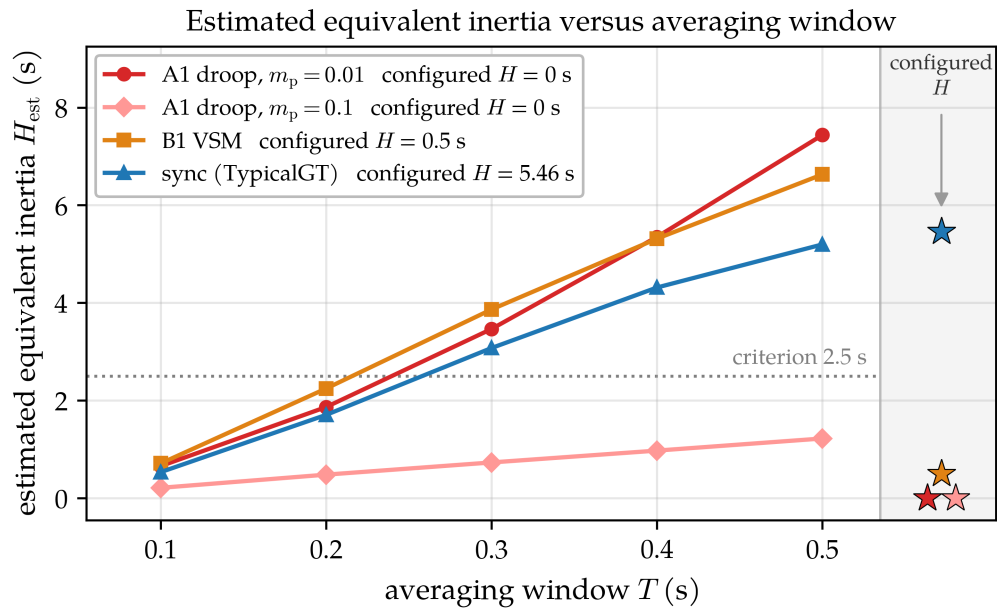


Figure 47. Estimated equivalent inertia versus integration-window length for the tested control configurations. The window dependence illustrates the contribution of damping-related power response. Configured virtual inertia and filtered-droop equivalent inertia are distinct quantities.

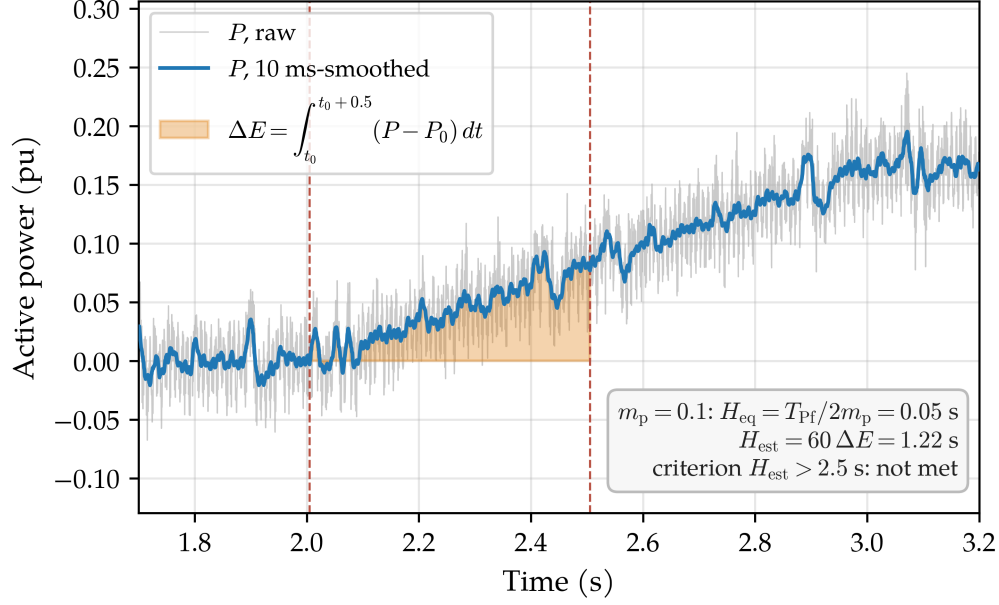


Figure 48. GFM inertia estimation test at $m_p = 0.1$: $H_{\text{est}} = 1.22$ s. The weaker droop reduces both equivalent damping and filtered-droop equivalent inertia.

5 Next Steps and Conclusion

The first-year studies establish a platform for examining how inverter controls affect system-level stability. They also motivate analytical models that relate control parameters to fault-duration limits. The next phase will pair corresponding PSS/E and PSCAD studies while extending the preliminary energy analysis described below.

5.1 Next Steps

We will align network topology, operating points, device ratings, bases, applicable control settings, and disturbance definitions across the positive-sequence and EMT models. The purpose is to understand which dynamics each domain retains and how those differences affect engineering conclusions. We will not force identical trajectories from models that represent different phenomena. The adapted PSS/E network and its E-TRAN translation provide the common starting point.

The GFM studies show recovery across the tested replacement configurations, but they do not yet explain every control parameter's influence on stability. We therefore seek analytical estimates of the maximum tolerable fault duration for systems containing multiple GFM resources. Such estimates could help manufacturers and operators interpret the effects of droop, current limits, and voltage limits. The following preliminary exploration applies familiar machine-based methods before pursuing more detailed formulations.

5.1.1 Preliminary Application of Classical Energy Methods

Classical energy-function methods offer a familiar bridge from synchronous-machine stability analysis to GFM research [22, 26, 27, 28, 29]. These methods compare an energy-like measure of the post-fault state with an energy boundary associated with unstable equilibria. They can provide sufficient stability conditions for the assumed model without following every subsequent swing. Applying them to GFM systems requires attention to the assumptions behind the power-angle

relationship and damping representation.

For the post-fault system with center-of-inertia states $(\theta, \tilde{\omega})$, the transient energy function is the first integral of the swing equations [22],

$$V(\theta, \tilde{\omega}) = \frac{1}{2} \sum_i M_i \tilde{\omega}_i^2 - \sum_i \int_{\theta_i^s}^{\theta_i} f_i(\theta) d\theta_i = V_{KE} + V_{PE}, \quad (16)$$

where f_i is the accelerating power of device i in the center-of-inertia frame and θ^s is the post-fault stable equilibrium. Written out for n devices represented as constant internal voltages E_i behind their coupling reactances, with the network reduced to those internal nodes [22],

$$f_i(\theta) = P_{m,i} - P_{e,i}(\theta) - \frac{M_i}{M_T} P_{COI}(\theta), \quad P_{e,i}(\theta) = E_i \sum_{j=1}^n E_j [G_{ij} \cos(\theta_i - \theta_j) + B_{ij} \sin(\theta_i - \theta_j)], \quad (17)$$

with $P_{COI} = \sum_i (P_{m,i} - P_{e,i})$ and $M_T = \sum_i M_i$. Here $P_{m,i}$ is the mechanical power of a machine or the power reference p^* of an inverter, and G_{ij} and B_{ij} are the conductance and susceptance of the reduced network between internal nodes i and j ; in the 39-bus cases the electrical power is taken from the constant-power network solution instead of a reduced admittance matrix, with the same dependence on the internal voltages and the coupling reactances. So f_i , and with it the potential energy V_{PE} and the shape of the surfaces in Figures 50 and 51, contain the dispatch, the internal voltages, and the network admittances only.

The inertia coefficients enter the kinetic term and the center-of-inertia weighting M_i/M_T , and the damping coefficients D_i do not appear in V at all. Damping acts only on how V changes along a trajectory. Without damping, $dV/dt = 0$ along post-fault trajectories, so a state with $V < V_{cr}$, the potential energy at the unstable equilibrium that bounds the basin, cannot reach that boundary: V is a Lyapunov function, that is, a function that is positive about the equilibrium and non-increasing along the system's trajectories [32], and $V < V_{cr}$ is a sufficient stability condition.

The machine estimates below use $D = 0$. For the droop inverters the damping $D_i = 1/m_p$ is retained; because every unit shares the same power-filter time constant, the ratio $D_i/M_i = 1/T_{PF}$ is the same on every unit, the center-of-inertia transform remains exact, and $dV/dt = -\sum_i D_i \tilde{\omega}_i^2 \leq 0$, so the same bound holds with the damping in place. For the single-machine model, $V_{cr} = V_{PE}(\delta_u)$, and the condition $V \leq V_{cr}$ at clearing is the equal-area criterion.

The critical clearing time follows in three steps: compute θ^s ; obtain V_{cr} from the unstable equilibrium or, in the potential-energy-boundary-surface (PEBS) screen, from the maximum of V_{PE} along the sustained-fault trajectory; integrate the fault-on equations from the pre-fault equilibrium and record the time at which V reaches V_{cr} . In this report each device is modeled as a constant voltage behind reactance, E' behind X'_d for the machines and the reduced droop model of Equation (6) for the inverters, with coefficients converted to the system base. The 39-bus network is retained with the model's constant-power loads, the fault is the bolted bus-14 fault with the network intact, and V_{PE} is evaluated as the path integral in Equation (16) along the computed trajectory. The quoted estimates are the bisection limits of the integrated reduced model, with stability defined by bounded post-clearing angles; the PEBS screen gives 0.228 s for the ten-machine case.

For the constant-voltage, lossless single-machine model in Equation (2), the stable and unstable equilibria satisfy

$$\delta_s = \arcsin(p_m/p_{\max}), \quad \delta_u = \pi - \delta_s, \quad (18)$$

for the positive-power operating branch with $0 < p_m < p_{\max}$. The unstable equilibrium helps define the energy boundary, but its angle alone is not a complete transient-stability criterion because speed

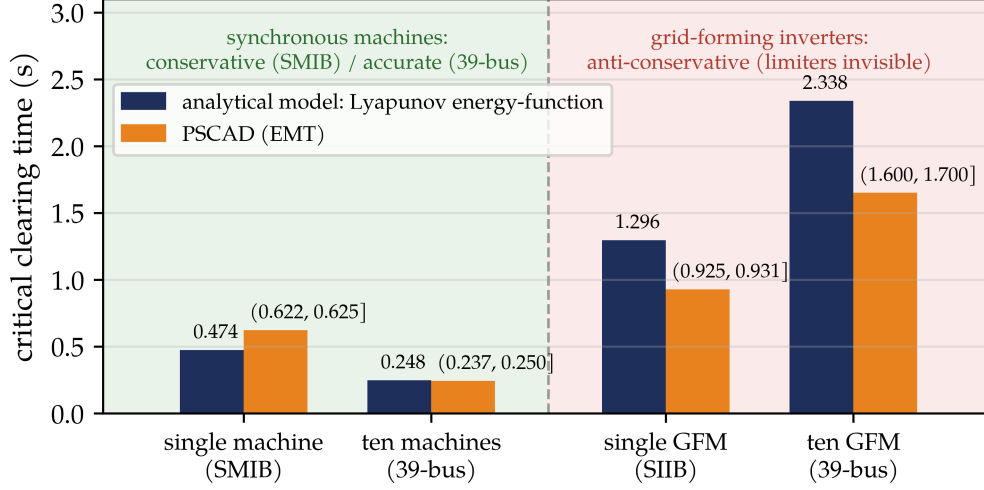


Figure 49. Energy-method critical clearing times (navy) compared with PSCAD EMT brackets (orange). Each bracket runs from a successful to an unsuccessful clearing duration; the estimates are single values from the reduced models of Section 5.1.1: the machine model is the undamped classical one, and the inverter model includes the droop damping $D = 1/m_p$. The inverter estimates exceed their brackets because those models carry no current limiter (Section 5.1.2).

also contributes. An energy estimate combines the fault-on trajectory and the post-fault boundary to estimate critical clearing time. That time is the maximum fault duration satisfying the selected post-fault stability criterion.

We compare analytical estimates with brackets $(a, b]$ formed by a successful simulation at duration a and an unsuccessful simulation at b . The angle-based comparisons distinguish return toward the original equilibrium from a phase slip into a neighboring angle region. A phase slip does not necessarily imply sustained loss of synchronism, and an unwrapped angle near 360° alone does not show whether the system later resynchronizes. These distinctions matter when interpreting critical clearing time as a practical ride-through limit.

Figure 49 summarizes the preliminary comparisons. The machine estimates hold the mechanical power constant and omit the exciter, governor, and stabilizer; the single-machine case is lossless with zero electrical power during the bolted fault. The GFM estimates use the swing-form droop model of Equation (5) with $H_{eq} = 0.5$ s and $D = 1/m_p = 100$, a constant internal voltage behind the filter and transformer reactance, and no current or voltage limit. For the single-machine case, the 0.474 s estimate is below the demonstrated stable duration of 0.622 s. For the 39-bus machine case, the 0.248 s estimate lies inside the $(0.237, 0.250]$ s simulation bracket. An estimate inside a bracket is consistent with its resolution but is not, by itself, demonstrably conservative.

Following the center-of-inertia (COI) formulation in Sauer and Pai, Chapter 9, Section 9.5, Equation (9.19) [22], define

$$\delta_{COI} = \frac{\sum_{i=1}^n M_i \delta_i}{\sum_{i=1}^n M_i}, \quad \theta_i = \delta_i - \delta_{COI}, \quad \tilde{\theta}_i = \theta_i - \theta_i^s. \quad (19)$$

Here M_i is the inertia coefficient in the reduced angular-acceleration equation, expressed on a common system base, and θ_i^s is the reference stable-equilibrium angle. For a synchronous machine, δ_i is the electrical rotor angle. For a GFM inverter, it is the controller-generated electrical angle; the analogous COI uses the reduced model's equivalent inertia coefficients, not physical rotor inertia. The plotted coordinates $\tilde{\theta}_{31}$ and $\tilde{\theta}_{32}$ are therefore equilibrium-shifted angles of the resources

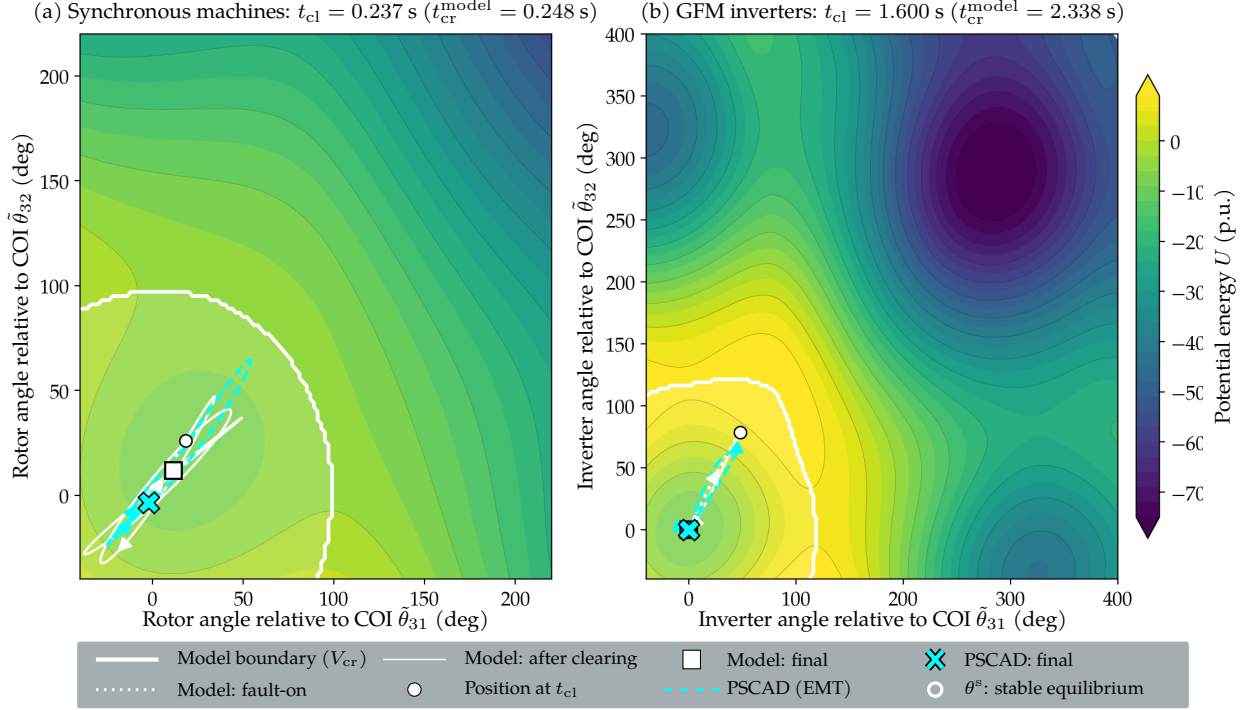


Figure 50. Potential-energy contours in equilibrium-shifted COI coordinates for the 39-bus system: (a) ten synchronous machines and (b) ten GFM inverters. Clearing durations are the last stable cases, 0.237 s and 1.600 s, respectively. White dotted and thin solid curves show the model's fault-on and post-clearing paths; cyan dashed curves show PSCAD EMT trajectories. The thick white curve is the model boundary estimate. Circles mark clearing, squares mark model endpoints, and cyan crosses mark EMT endpoints. The colour scale is shared by both panels.

at buses 31 and 32 relative to the fleet's COI, not pairwise inter-machine angles. The reference equilibrium appears at the origin.

Figure 50 shows the machine and GFM trajectories at their last stable clearing durations, 0.237 s and 1.6 s, respectively. The two-dimensional (2D) contours display analytical potential energy over these COI coordinates. Both responses remain near the reference equilibrium after clearing, although the marked model and EMT endpoints need not coincide. The white boundary is the reduced model's critical-energy boundary estimate; a two-angle view does not certify a basin boundary for the complete EMT system, whose state also includes speeds and other control variables.

5.1.2 Limits of the Present Energy Construction

The same classical construction overestimates the clearing margin for the tested GFM cases. In the 39-bus comparison, it predicts approximately 2.34 s against an EMT bracket of (1.6, 1.7] s. In the single-inverter infinite-bus comparison, it predicts 1.296 s against (0.925, 0.931] s. These differences are larger than in the corresponding machine comparisons and motivate a more detailed treatment of inverter controls.

Figure 51 compares clearing near each analytical critical clearing time using complementary 2D and three-dimensional (3D) views of the same cases. The contours make the trajectories and endpoints easier to compare, while the 3D surfaces expose the potential-energy ridge and neighboring well. At 0.2455 s, the machine EMT trajectory returns near the reference equilibrium.

At 2.315 s, the GFM EMT trajectory crosses the ridge and ends near $(\tilde{\theta}_{31}, \tilde{\theta}_{32}) = (290^\circ, 290^\circ)$ in the supplied COI-coordinate view, while the analytical trajectory returns toward the origin. The trajectories disagree on return to the original equilibrium, which is a more specific observation than a claim that the inverter never resynchronizes.

The larger amplitude and curvature of the GFM surface follow from the coupling reactances, not from the inertia or damping coefficients: neither appears in f_i of Equation (17), so neither enters the potential energy of Equation (16). Each inverter is represented as E behind $X_L = 0.15$ per unit on its base $S_{\text{inv,base}} = P_G/0.6$, the example value of the model specification [10], whereas each machine is E' behind $X'_d = 0.26$ per unit on its rating; on the system base the inverter internal nodes are coupled through 2.8–2.9 times less reactance at nine of the ten buses, and the curvature of the surface at the equilibrium is 1.9 times the machine value. Recomputing the inverter surface with the machines' system-base reactances brings that curvature to within 25% of the machine value.

Current limiting and the ceiling on commanded voltage are plausible contributors to this discrepancy. When these constraints become active, the realized electrical power need not follow the sinusoidal relationship used by the classical construction. Figure 53 illustrates how saturation changes the power-angle characteristic and the associated unstable equilibrium. Future controlled studies will isolate these effects rather than assume that omitted limiters explain every difference. The issue is the applicability of the chosen model assumptions, not a violation of classical stability theory. Figure 52 shows the measured single-inverter clearing-time bracket as the transient current limit is raised from 2.0 to 15 per unit with all other settings fixed: the bracket rises from $(0.925, 0.931]$ s at 2.0 per unit to $(1.163, 1.169]$ s at 12 per unit and is unchanged at 15 per unit, where the limit exceeds the unlimited fault-current demand.

5.1.3 Modeling Goals

The next analytical steps build on this preliminary comparison. We will connect the model extensions to the same disturbances used in the testbed. Hardware studies remain a future direction after the simulation comparisons establish appropriate test cases.

- **Limiter-aware stability estimates.** Develop a critical-clearing-time analytical model for single and multi-inverter systems to include saturation and determine conditions under which the estimate is conservative.
- **Guidance for manufacturers and operators.** Identify control parameters with the greatest influence on system-level stability for the configurations studied.
- **Mixed-generation systems.** Extend the analysis to combinations of synchronous machines, GFM inverters of various control approaches, and GFL inverters.
- **Additional disturbances.** Evaluate the models under unbalanced faults at other locations, faults with nonzero impedance, phase-angle jumps, and load steps.
- **Large electronic loads.** Integrate large electronic load profiles and evaluate their effect on the fault ride-through capability of a system with a high share of grid-forming inverters.
- **Hardware validation.** Investigate single-inverter infinite-bus and hardware-in-the-loop tests, followed by multi-inverter disturbances in the UT Austin laboratory microgrid.

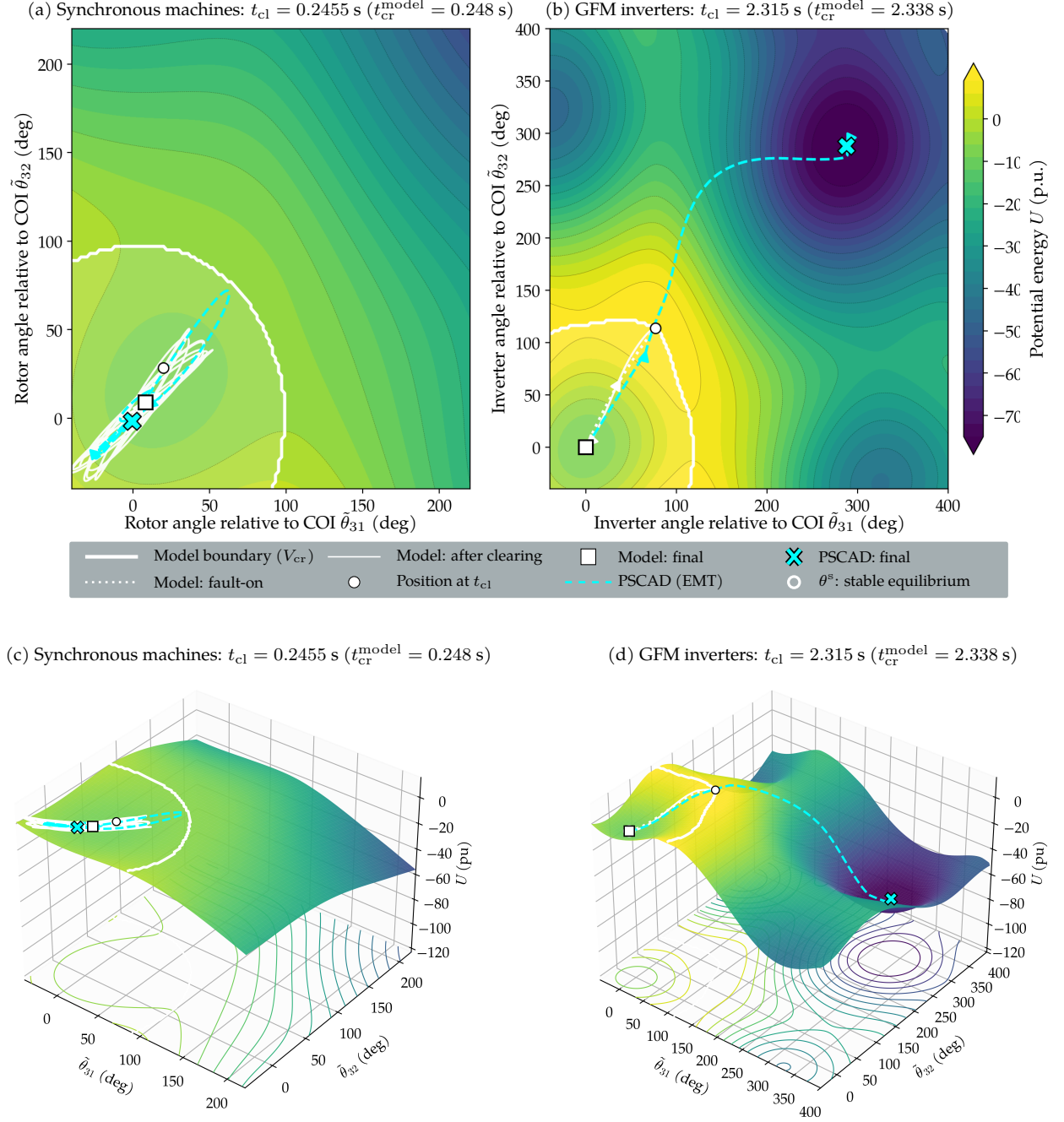


Figure 51. Complementary views near the analytical critical clearing times: (a,b) 2D potential-energy contours and (c,d) 3D surfaces, with synchronous machines on the left and GFM inverters on the right. Clearing durations are 0.2455 s and 2.315 s; model estimates are 0.248 s and 2.338 s. All panels use the equilibrium-shifted COI angles in Equation (19) and the shared legend above the 3D views. The GFM EMT endpoint lies in a neighboring well, unlike the model endpoint near the origin. EMT clearing-time brackets remain $(0.237, 0.250]$ s and $(1.600, 1.700]$ s.

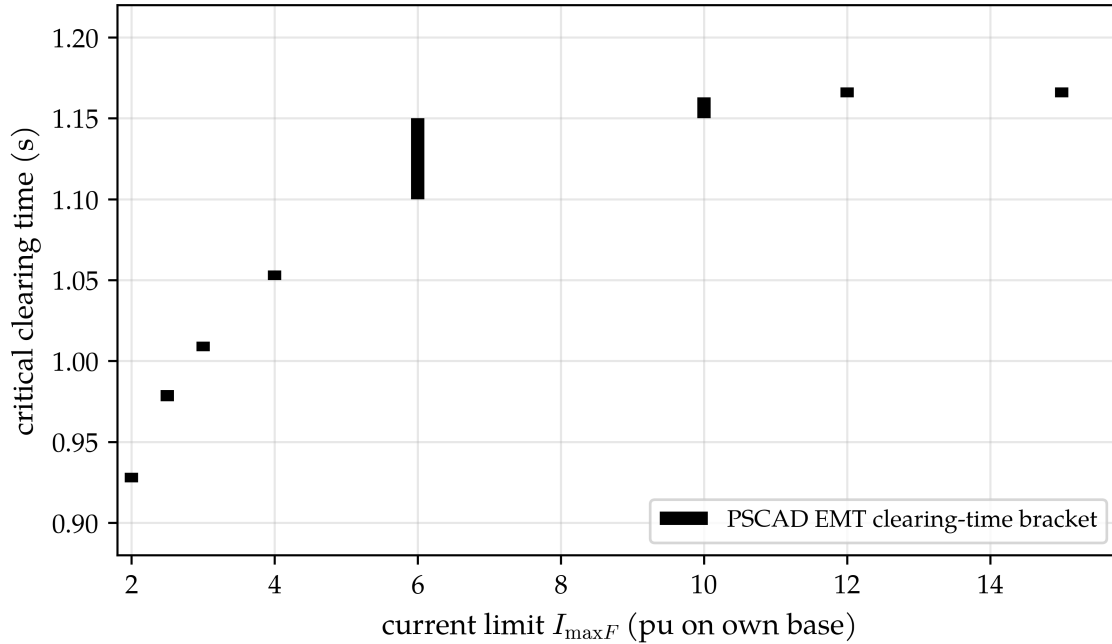


Figure 52. Measured critical clearing time of the single inverter on the infinite bus as a function of the transient current limit $I_{\max F}$, all other settings fixed. Each bar is the bracket between the longest stable and the shortest unstable fault duration.

5.2 Conclusion

We developed a reusable PSCAD EMT testbed for studying networks with increasing numbers of GFM resources. Starting from an IEEE 39-bus PSS/E case, we used E-TRAN to construct the network and equipped it with representative machine dynamics and selected inverter models. The first-year studies show fault ride through across the tested replacement configurations and useful differences between machine and GFM responses. The selected droop controls provide strong equivalent damping and low equivalent inertia, which help interpret the faster settling observed in

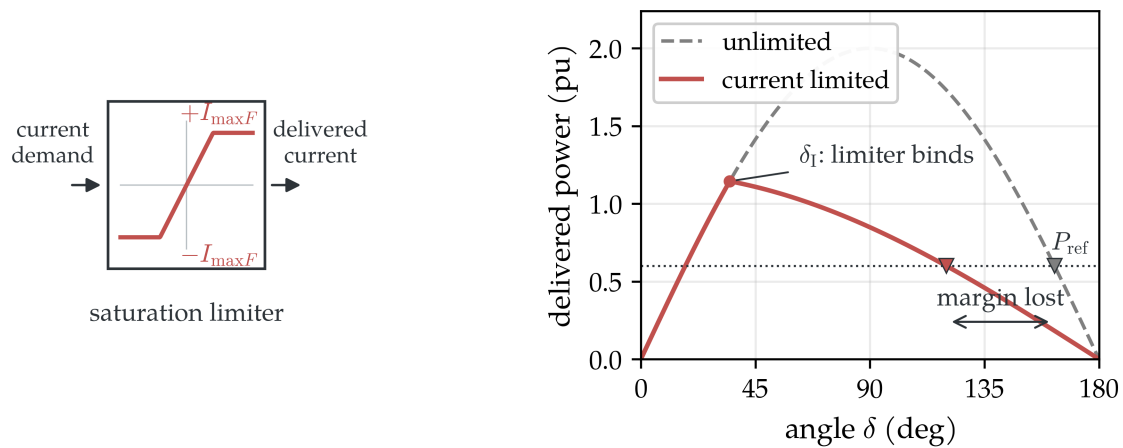


Figure 53. Conceptual effect of current saturation on the power-angle characteristic. The limited characteristic departs from the sinusoid, changing the location of the unstable equilibrium and the region of attraction represented by the classical construction.

the simulations.

The inertia-estimation and preliminary energy-method investigations connect those observations to familiar power-system concepts. They also show why damping contributions and active constraints matter when interpreting reduced models. The next phase will establish corresponding positive-sequence and EMT comparisons and examine how control, network, and load dynamics affect stability conclusions. The testbed provides a strong foundation for that work and for developing practical guidance on inverter-dominated power systems.

Acknowledgments

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A Synchronous Machine Model Parameters

The synchronous baseline combines GENROU, ESST4B, GGOV1, and PSS2B with an inertia constant of 5.46 s. The ten machines use the same ERCOT-supplied dynamic parameter set, while their ratings and operating points can differ. The bus-30 PSS/E record lists the parameters in standard model order:

```
30, 'GENROU', 1,
  7.61, 0.04, 0.98, 0.06, 5.46, 0, 1.99, 1.81,
  0.26, 0.45, 0.2, 0.15, 0.06, 0.32, //
30, 'ESST4B', 1,
  0, 4.0, 4.0, 1, -0.90, 0.01, 1, 0, 1, -0.90,
  0, 5.5, 0, 6.0, 0.1, 0, 0, //
30, 'GGOV1', 1,
  1, 1, 0.05, 1., 0.055, -0.055, 10, 1, 0, 1, 1,
  0.16, 0.5, 1.5, 0.2, 0.1, 0, 0, 3, 2, 0.7, 1,
  0, 0.1, -0.1, 0, 0.1, 10, 0.1, 0, 0.0002, 4, 5, 99, -99, //
30, 'PSS2B', 1,
  1, 0, 3, 0, 5, 1, 2, 2, 0, 2, 0, 2, 0.25, 1,
  0.5, 0.1, 15, 0.2, 0.031, 0.1, 0.02, 0, 0, 99, -99,
  99, -99, 0.1, -0.1, //
```

GENROU field 5 specifies the inertia constant $H = 5.46$ s. For the REGFM_A1 reference configuration, $m_p = 0.01$ and $T_{Pf} = 10$ ms give $H_{eq} = 0.5$ s and $D_{eq} = 100$ through Equation (5). The replacement sizing uses $S_{inv,base} = P_G/0.6$, where P_G is generator dispatch in MW and the resulting base is in MVA. The bus-14 reference fault is bolted three-phase-to-ground for five cycles without reclosing; its 2.0 per-unit current limit is distinct from the 1.5 per-unit bus-10 study.

B PSCAD Output Files

PSCAD writes numerical channels and channel descriptions to separate files. The processing workflow reconstructs them into a common table for analysis and plotting. This appendix describes that workflow, while the report also uses circuit diagrams, screenshots, and analytical illustrations.

B.1 What PSCAD Writes

For a project named `name`, EMTDC writes channel descriptions to `name.inf`. Numerical files follow the sequence `name_01.out`, `name_02.out`, and so on. Each output file contains simulation time followed by up to ten channels. The `PGB(...)` records in the `.inf` file provide channel indices and `Desc="..."` descriptions. The numerical files themselves do not name their columns.

B.2 Channel Naming

The harness parses the channel descriptions and converts them into column names. Repeated descriptions receive a suffix to keep the names unique. The following function illustrates this naming step:

```
def inf_channel_names():
    names = []
    for line in open(project + ".inf"):
        if line.startswith("PGB("):
            names.append(line.split('Desc="')[1].split('"')[0]
                          .replace(":", "_"))

    seen, out = {}, []
    for n in names:
        # disambiguate duplicates
        seen[n] = seen.get(n, 0) + 1
        out.append(n if seen[n] == 1 else "%s_dup%d" % (n, seen[n]))
    return out
```

B.3 Stitching and Export

The exporter combines the output-file family using `OutFile` from the PSCAD file utilities. It checks modification times against the run start to avoid reading a previous run's output and compares the column count with the channel descriptions. It then downsamples the table and selects the declared channels for export:

```
fout = OutFile(project_stem) # name_01.out, name_02.out, ...
fout.toCSV(csv="_res_full.csv") # stitched: TIME + all channels
df = pd.read_csv("_res_full.csv")
names = inf_channel_names()
if len(df.columns) != len(names) + 1: # TIME + channels, exactly
    raise RuntimeError("column mismatch vs .inf")
df.columns = ["TIME"] + names
df = df.loc[:, DOWNSAMPLE, ["TIME"] + KEEP]
df.to_csv(run_dir + "/rung_%s.csv" % tag, index=False)
```

Each run produces a CSV identified by its fault duration. This organization supports comparison of successful and unsuccessful runs when constructing clearing-time brackets. The channel names accompany the numerical values so that subsequent analysis can identify each recorded signal.

B.4 Plotting Conventions

Time-domain plots mark the fault interval and use a pre-fault window to establish the baseline. Network plots show RMS bus voltages in kV, currents in kA, and powers in MW and MVar; device comparisons use per unit on each device base, stated in the caption. Two current measurements appear: the generator-bus channels are taken on the 230 kV side of each unit's step-up transformer, and a GFM unit's internal current magnitude is the inverter's own measurement at its terminals on the 10 kV side, in per unit of the unit rating; each caption states which is shown. Axes identify

units, and multi-device panels share limits where appropriate. Ratings, dispatch, and terminal measurement locations remain important when comparing devices; per-unit presentation alone does not make the cases identical.

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